

POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Clustering and preferential sampling in homogeneous shear turbulence

Tesi di Laurea Magistrale in
Aeronautical Engineering - Ingegneria Aeronautica

Marco Zappoli, 243319

Advisor:
Dr. Alessandro Chiarini

Co-advisors:
Prof. Maurizio Quadrio

Academic year:
2024-2025

Abstract: Direct numerical simulations of inertial-particle dynamics in homogeneous shear turbulence are performed to investigate particle clustering and the mechanisms underlying preferential concentration. Particle motion is described by the full Maxey–Riley–Gatignol equation, including the Basset history force, enabling an explicit assessment of memory effects. Simulations are conducted for Stokes numbers $St = [0.1, 0.5, 1]$ and particle-to-fluid density ratios $\rho_p/\rho_f = [10, 100]$, allowing systematic exploration of inertia- and density-dependent behavior. Clustering intensity increases with particle inertia and is predominantly localized at dissipative scales, where particles preferentially sample strain-dominated regions while avoiding strong vortical structures. Mean shear induces pronounced anisotropy in particle distributions and cluster orientation, yet does not inhibit clustering or its multi-scale character. Voronoï analysis reveals self-similar cluster size distributions exhibiting power-law scaling and nearly constant internal particle concentration across a broad range of cluster sizes, indicating scale-invariant aggregation dynamics. Cluster morphology is characterized by elongated shapes and displays a high degree of universality, with only weak dependence on Stokes number and density ratio, consistent with previous findings in homogeneous isotropic turbulence. The inclusion of the Basset history force systematically reduces clustering intensity, smooths cluster geometry, and alters particle penetration into high-enstrophy regions. These results demonstrate that inertial-particle clustering in homogeneous shear turbulence arises from the interplay between dissipative-scale dynamics, shear-induced anisotropy, and non-local memory effects, leading to robust yet strongly structured particle organization across scales.

Key-words: Turbulence, Particles, Basset history force, Clustering, Preferential sampling

1. Introduction

The transport of inertial particles in turbulent flows plays a central role in a wide range of natural and engineering processes, including plankton encounters and ecological interactions in the ocean (Károlyi *et al.*, 2000; Durham & Stocker, 2012; De Lillo *et al.*, 2014; Niazi Ardekani *et al.*, 2017), droplet growth and collision dynamics in clouds (Falkovich *et al.*, 2002; Grabowski & Wang, 2013; Ravichandran & Govindarajan, 2017), dust aggregation and the early stages of planetesimal formation in protoplanetary disks (Bracco *et al.*, 1999; Johansen *et al.*, 2007; Chambers, 2010; Kleeorin & Rogachevskii, 2025), as well as spray formation and combustion processes (Domingo *et al.*, 2005). In all these systems, a turbulent carrier flow interacts with a dispersed solid or

liquid phase, producing highly non-uniform particle distributions. These spatial inhomogeneities have profound implications for collision rates, mixing efficiency, and effective transport properties (Balachandar & Eaton, 2010; Capecelatro *et al.*, 2015). A key challenge is therefore to understand where, and under which conditions, particles preferentially accumulate and to identify the physical mechanisms responsible for this behavior. In this work, we focus on inertial-particle clustering in homogeneous shear turbulence (HST), aiming to isolate the influence of mean shear on clustering mechanisms and anisotropic spatial distributions.

Unlike passive Lagrangian tracers, inertial particles do not follow the fluid flow exactly. Their finite inertia causes trajectories to deviate from the local fluid motion, leading to complex transport dynamics governed by the interplay between hydrodynamic forces and the multiscale structure of turbulence. A prominent manifestation of this interplay is preferential concentration, or clustering, whereby particles accumulate unevenly in space (Squires & Eaton, 1991; Eaton & Fessler, 1994). This phenomenon produces particle-rich regions (clusters) and particle-depleted regions (voids), which significantly alter the statistical and dynamical properties of particle-laden flows. In particular, inertial particles tend to sample specific subsets of the carrier-flow structures rather than exploring the flow uniformly. Several physical mechanisms have been proposed to explain this behavior. Centrifugal ejection from vortical structures (Maxey, 1987), preferential accumulation in high-strain regions, and the sweep-stick mechanism (Chen *et al.*, 2006; Goto & Vassilicos, 2006, 2008; Coleman & Vassilicos, 2009) are among the most widely recognized. Collectively, these mechanisms determine both the intensity and morphology of particle clusters, shaping their spatial organization and dynamic interactions with the turbulent flow.

Much of the fundamental understanding of inertial-particle clustering comes from studies in homogeneous isotropic turbulence (HIT), which provides a canonical framework for exploring preferential sampling due to its statistical simplicity and well-characterized properties (Wang & Maxey, 1993; Balkovsky *et al.*, 2001; Aliseda *et al.*, 2002; Bec *et al.*, 2007; Ireland *et al.*, 2016). These studies typically assume dilute suspensions of particles that are small compared to the smallest turbulent scales, allowing the use of the point-particle approximation. The validity of this approximation has been assessed against fully resolved simulations, such as those employing immersed boundary methods (e.g. Uhlmann & Chouippe, 2017; Chiarini & Rosti, 2024). In this study we adopt the point-particle approximation, which keeps computational costs manageable while enabling the investigation of large particle ensembles. Within the point-particle framework, different levels of coupling between particles and fluid can be considered. In one-way coupling, particles are passively advected by the flow and do not influence it. In two-way coupling, momentum exchange between particles and fluid is taken into account (Elghobashi & Truesdell, 1993). In four-way coupling, particle-particle collisions are also included, ensuring excluded-volume effects and preventing unphysical clustering (Bec *et al.*, 2013). Because we consider dilute conditions, where both the particle back-reaction on the carrier flow and inter-particle collisions are negligible, we adopt the one-way coupled point-particle approximation.

Despite the fundamental insights gained from studies in HIT, real turbulent flows are rarely homogeneous or isotropic. Mean shear, rotation, buoyancy, and confinement introduce directional biases and spatial non-uniformities, which can fundamentally alter clustering mechanisms. Wall-bounded flows, such as channels and boundary layers, are canonical examples of such anisotropic and inhomogeneous turbulence. In these configurations, additional transport mechanisms arise, most notably turbophoresis, which drives particles toward regions of reduced turbulent kinetic energy near walls (Reeks, 1983; Rouson & Eaton, 2001; Portela *et al.*, 2002; Righetti & Romano, 2004), while coherent near-wall structures (streaks, quasi-streamwise vortices, sweeps, and ejections) strongly influence particle dynamics. Disentangling the effects of mean shear from wall and inhomogeneity effects, however, is challenging. Homogeneous shear turbulence offers a physically relevant and controlled setting to study anisotropy in isolation. By retaining a mean shear while preserving spatial homogeneity and eliminating walls, HST allows the direct investigation of how shear-induced anisotropy affects particle dynamics.

Previous studies of inertial particles in HST have investigated clustering and anisotropy using simplified point-particle models in which the Basset history force is neglected, retaining only a reduced form of the Maxey–Riley–Gatignol (MRG) equations (Maxey & Riley, 1983; Gatignol, 1983). Within this framework, Gualtieri *et al.* (2009) showed, through the analysis of the angular distribution function, that mean shear can induce preferential orientations in particle distributions, even at scales where the carrier flow appears nearly isotropic. These results were later confirmed experimentally by Nicolai *et al.* (2014) and further extended by Lee & Lee (2020) using two-dimensional distribution functions and principal-component analyses for settling particles. Despite these advances, preferential sampling and clustering in HST remain far from fully understood. In particular, it is still unclear how shear-induced anisotropy affects the topology and geometry of particle clusters across scales, whether the preferential sampling mechanisms identified in HIT persist or are fundamentally altered in the presence of mean shear, and how these effects depend on particle inertia. Moreover, all existing numerical HST studies to date neglect the Basset history force, which accounts for the unsteady diffusion of vorticity in the particle boundary layer and introduces a non-local time convolution over the particle's past motion. While often omitted due to its substantial computational and memory cost, recent investigations in HIT have demonstrated that the Basset history force can contribute a non-negligible fraction of the particle acceleration (particularly for particles with sizes on the order of the Kolmogorov length scale) and can significantly modify clustering

statistics (Olivieri *et al.*, 2014; Daitche, 2015). The absence of this force in previous HST studies therefore leaves open fundamental questions regarding its impact on anisotropic clustering, preferential sampling, and the validity of reduced-force models in shear-dominated turbulent flows.
The present work aims to address these gaps through a systematic investigation of inertial-particle clustering and preferential sampling in HST. Specifically the objectives of the present study are fourfold: *(i)* to quantify the relative contributions of the individual force terms in the MRG equation in anisotropic turbulent environments, thereby extending the analyses of Olivieri *et al.* (2014) and Daitche (2015) beyond HIT and identifying the conditions under which simplified MRG models remain appropriate; *(ii)* to assess the impact of the Basset history force on small-scale clustering intensity and anisotropic features, extending previous HST studies based on reduced force models (Gualtieri *et al.*, 2009); *(iii)* to characterize the geometric and topological properties of inertial-particle clusters in HST, extending Voronoï-based analyses previously performed in HIT (Baker *et al.*, 2017); and *(iv)* to elucidate the physical mechanisms governing inertial-particle clustering in anisotropic turbulence by relating particle accumulation patterns to the velocity and acceleration fields sampled by the particles, within the frameworks of both the centrifugal and sweep-stick mechanisms.
The remainder of the paper is organized as follows. Section 2 presents the computational setup and numerical methods for both the carrier flow and particle dynamics. Section 3 provides a statistical characterization of HST. In Section 4, we analyze the relative importance of the different MRG force terms and their role in particle dynamics. Sections 5 and 6 introduce clustering analyses based on radial and angular distribution functions and Voronoï tessellation, examining intensity, orientation, and cluster geometry. Finally, Section 7 investigates preferential sampling, interpreting particle accumulation in the context of centrifugal and sweep-stick mechanisms.

2. Problem formulation and numerical methods

In this section, we describe the formulation of the HST configuration, the numerical strategy adopted for the direct numerical simulation (DNS) of the carrier flow, and the Lagrangian framework used for particle tracking. Particular emphasis is placed on the consistent treatment of shear-induced anisotropy, the implementation of shear-periodic boundary conditions, and the accurate integration of the complete MRG equation.

2.1. Carrier flow

The carrier flow velocity and vorticity are decomposed into their ensemble means and fluctuations as:

$$\boldsymbol{U}(\boldsymbol{x},t) = \boldsymbol{U_0}(z) + \boldsymbol{u}(\boldsymbol{x},t), \qquad \boldsymbol{\Omega}(\boldsymbol{x},t) = \nabla \times \boldsymbol{U}(\boldsymbol{x},t) = \boldsymbol{\Omega_0} + \boldsymbol{\omega}(\boldsymbol{x},t),$$

where the ensemble averages (implemented as spatial and temporal averaging) are:

$$\boldsymbol{U_0}(z) = \langle \boldsymbol{U}(\boldsymbol{x},t) \rangle = (Sz, 0, 0), \qquad \boldsymbol{\Omega_0} = \langle \boldsymbol{\Omega}(\boldsymbol{x},t) \rangle = (0, S, 0).$$

Here the Cartesian coordinate system is chosen such that x, y and z denote the streamwise, spanwise and vertical directions, respectively, with corresponding fluctuating velocity components (u, v, w) and vorticity components (ξ, η, ζ).
Following the classical reduction adopted in homogeneous shear flows (Kim *et al.*, 1987), the incompressible Navier–Stokes equations governing the fluctuating field are recast in terms of two scalar variables: the Laplacian of the vertical velocity, $\phi(\boldsymbol{x},t) \equiv \nabla^2 w(\boldsymbol{x},t)$, and the vertical component of the vorticity, $\zeta(\boldsymbol{x},t)$. The advection by the mean shear is written explicitly, so that the material derivative is split as $\partial_t + Sz\,\partial_x$.
The resulting evolution equations in the velocity–vorticity formulation read

$$\frac{\partial \zeta}{\partial t} + Sz\frac{\partial \zeta}{\partial x} = h_\zeta + \nu\nabla^2\zeta, \qquad \frac{\partial \phi}{\partial t} + Sz\frac{\partial \phi}{\partial x} = h_w + \nu\nabla^2\phi. \tag{1}$$

where ν is the kinematic viscosity. These two scalar equations are fully equivalent to the incompressible Navier–Stokes system, provided that the continuity constraint is enforced and appropriate shear-periodic boundary conditions are applied.
The nonlinear forcing terms are written compactly using $\boldsymbol{H} = \boldsymbol{u} \times \boldsymbol{\omega}$ as:

$$h_\zeta \equiv \frac{\partial H_x}{\partial y} - \frac{\partial H_y}{\partial x} - S\frac{\partial w}{\partial y}, \qquad h_w \equiv -\frac{\partial}{\partial z}\left(\frac{\partial H_x}{\partial x} + \frac{\partial H_y}{\partial y}\right) + \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) H_z. \tag{2}$$

The additional term proportional to S in h_ζ reflects the direct coupling between the mean shear and the fluctuating vertical velocity, and represents a key source of anisotropy in the vorticity dynamics.
In addition, the governing equations for the plane-averaged velocity components $\langle \boldsymbol{u} \rangle_{xy}$ are:

$$\frac{\partial\langle u\rangle_{xy}}{\partial t} = -\frac{\partial\langle uw\rangle_{xy}}{\partial z} + \nu\frac{\partial^2\langle u\rangle_{xy}}{\partial z^2}, \qquad \frac{\partial\langle v\rangle_{xy}}{\partial t} = -\frac{\partial\langle vw\rangle_{xy}}{\partial z} + \nu\frac{\partial^2\langle v\rangle_{xy}}{\partial z^2}, \tag{3}$$

where $\langle uw\rangle_{xy}$ and $\langle vw\rangle_{xy}$ represent the plane-averaged vertical fluxes of horizontal momentum. Continuity requires $\partial\langle w\rangle_{xy}/\partial z = 0$, hence $\langle w\rangle_{xy} = 0$.

2.2. Numerical method

The simulations are performed using a non-shearing reference formulation (Sekimoto *et al.*, 2016), in which the equations are solved in laboratory coordinates and the effect of the mean shear is treated analytically. Fourier expansions are employed in the homogeneous directions (x and y), while the vertical direction (z) is discretized using compact finite differences.
The numerical code is based on a modified version of the channel-flow solver developed by Luchini & Quadrio (2006), written in the CPL programming language (Luchini, 2020). Spatial and temporal discretizations follow the schemes described therein; only the aspects specific to HST and to particle tracking are detailed below.

2.2.1 Computational domain and discretization

Although HST is, in principle, unbounded, the direct numerical simulations are carried out in a finite rectangular computational domain of size $L_x \times L_y \times L_z$. Periodic boundary conditions are applied in the streamwise and spanwise directions, while shear-periodic boundary conditions are imposed in the vertical direction in order to properly account for the continuous mean displacement induced by the imposed shear.
Spatial discretization is based on Fourier spectral expansions in the homogeneous directions, providing spectral accuracy in the representation of turbulent velocity fluctuations. In the vertical direction, spatial derivatives are approximated using Padé-type compact finite-difference schemes on a uniform grid with N_z collocation points; specifically, a five-point stencil with fourth-order accuracy is employed.
Nonlinear terms are evaluated using a pseudo-spectral formulation and de-aliased through the standard 3/2-rule to prevent spurious energy transfer among resolved scales.

2.2.2 Shear-periodic boundary conditions

In HST, periodicity in the vertical direction is preserved only up to a streamwise shift proportional to the mean shear. In spectral space, the mapping between periodic images is given by

$$\widehat{g}(k_x, k_y, z, t) = \widehat{g}(k_x, k_y, z + nL_z, t)\, \exp\big(ik_x nSL_z t\big), \tag{4}$$

which reflects the continuous advection of fluid structures by the background shear.
This condition is enforced via a compact scheme in matrix form $BF^{(n)} = AF$, where A and B are penta-diagonal matrices with complex wrap-around entries. For even derivatives, B takes the form:

$$B = \begin{bmatrix} \delta & \alpha & \beta & 0 & \dots & 0 & \beta'^* & \alpha'^* \\ \alpha & \delta & \alpha & \beta & 0 & \dots & 0 & \beta'^* \\ \beta & \alpha & \delta & \alpha & \beta & 0 & \dots & 0 \\ \vdots & & \ddots & & & \ddots & & \vdots \\ 0 & \dots & 0 & \beta & \alpha & \delta & \alpha & \beta \\ \beta' & 0 & \dots & 0 & \beta & \alpha & \delta & \alpha \\ \alpha' & \beta' & 0 & \dots & 0 & \beta & \alpha & \delta \end{bmatrix}, \tag{5}$$

where $\alpha, \beta, \delta \in \mathbb{R}$ are the interior stencil coefficients and $\alpha', \beta' \in \mathbb{C}$ encode the phase shift across the periodic image. The starred entries are the complex conjugates. The wrap-around terms are the real coefficients multiplied by the appropriate phase factor mandated by equation (4); for instance

$$\alpha' = \alpha\, \mathrm{e}^{-ik_x SL_z t}, \qquad \alpha'^* = \alpha\, \mathrm{e}^{ik_x SL_z t}. \tag{6}$$

Because the top boundary is shifted relative to the bottom boundary, a *box period* T_s can be defined as the time required for the shifted image to realign with the original box:

$$T_s = \frac{L_x}{SL_z}, \qquad \text{hence} \qquad ST_s = \frac{L_x}{L_z}. \tag{7}$$

2.2.3 Time integration and analytical treatment of mean shear

The linear advection associated with the mean shear imposes severe stability constraints if treated explicitly. To overcome this limitation, the shear contribution is analytically absorbed through a time-dependent phase transformation in Fourier space,

$$\widetilde{f} \equiv \mathrm{e}^{\mathrm{i}k_x Szt}\,\widehat{f},$$

where $\widehat{f}$ denotes a generic Fourier coefficient. This transformation removes the mean-shear term from the time derivative and allows the remaining dynamics to be advanced using standard time-stepping schemes.
The equations are integrated using a mixed implicit–explicit strategy: nonlinear terms are advanced explicitly with a three-stage low-storage Runge–Kutta scheme, while viscous terms are treated implicitly using a second-order Crank–Nicolson discretization. This approach ensures numerical stability without compromising accuracy. To preserve the benefit of the analytical shear absorption, each Runge–Kutta sub-step includes a sequence of phase rotations that map intermediate fields between different stage reference frames. The resulting algorithm efficiently decouples the mean shear from the CFL constraint while retaining full spectral accuracy.
For the vertical velocity, the discretized sub-step reads:

$$\begin{aligned}\frac{\lambda_n}{\Delta t}\mathcal{E}_n[D_2(\widehat{w}^n)-k^2\widehat{w}^n]-\frac{\mathcal{E}_n}{Re}[D_4(\widehat{w}^n)-2k^2D_2(\widehat{w}^n)+k^4\widehat{w}^n]=\\ \frac{\lambda_n}{\Delta t}\mathcal{E}_n[D_2(\widehat{w}^{n-1})-k^2\widehat{w}^{n-1}]+\frac{1}{Re}[D_4(\widehat{w}^{n-1})-2k^2D_2(\widehat{w}^{n-1})+k^4\widehat{w}^{n-1}]+\theta_n\mathcal{N}^n+\xi_n\mathcal{E}_{n-1}\mathcal{N}^{n-1},\end{aligned} \tag{8}$$

where $\mathcal{E}_n = \exp[ik_xSz(c_n-c_{n-1})\Delta t]$ and $\mathcal{N}$ represents the nonlinear terms. The low-storage coefficients $\{\lambda_n, c_n, \theta_n, \xi_n\}$ ensure optimal stability and reduced memory overhead:

$$\lambda_n=\left\{\frac{120}{32},\frac{120}{8},\frac{120}{20}\right\},\qquad c_n=\left\{\frac{2}{\lambda_1},\frac{2}{\lambda_2},\frac{2}{\lambda_3}\right\},\qquad \theta_n=\left\{2,\frac{50}{8},\frac{90}{20}\right\},\qquad \xi_n=\left\{0,-\frac{34}{8},-\frac{50}{20}\right\}. \tag{9}$$

2.3. One-way point-particle simulations

Simulations are performed in the one-way coupling regime, appropriate for dilute suspensions, where particles are influenced by the flow but do not exert feedback on it and particle-particle interactions are neglected.
This modelling choice is fully consistent with the objectives of the present work, which focuses on how shear-induced anisotropy and turbulent fluctuations modulate particle clustering, preferential concentration and sampling.
Each particle is assumed to be spherical and is governed by the complete Maxey–Riley–Gatignol equation (Maxey & Riley, 1983; Gatignol, 1983). Denoting by ρ_p the particle density, V_p its volume, d_p its diameter, and by ρ_f and ν the density and kinematic viscosity of the fluid, respectively, the particle momentum equation reads

$$\begin{aligned}\rho_pV_p\frac{d\boldsymbol{U_p}}{dt} &= \underbrace{3\pi d_p\rho_f\nu\left(\boldsymbol{U}-\boldsymbol{U_p}+\frac{1}{6}\left(\frac{d_p}{2}\right)^2\nabla^2\boldsymbol{U}\right)}_{\textit{Stokes drag}}+\underbrace{\rho_fV_p\frac{D\boldsymbol{U}}{Dt}}_{\textit{Pressure gradient}}\\ &+\underbrace{\frac{\rho_fV_p}{2}\left(\frac{D\boldsymbol{U}}{Dt}-\frac{d\boldsymbol{U_p}}{dt}+\frac{1}{10}\left(\frac{d_p}{2}\right)^2\frac{d}{dt}(\nabla^2\boldsymbol{U})\right)}_{\textit{Added mass}}\\ &+\underbrace{\frac{3}{2}d_p^2\rho_f\sqrt{\pi\nu}\int_{-\infty}^{t}K_B(t-\tau)\left(\frac{d\boldsymbol{U}}{d\tau}-\frac{d\boldsymbol{U_p}}{d\tau}+\frac{1}{6}\left(\frac{d_p}{2}\right)^2\frac{d}{d\tau}\left(\nabla^2\boldsymbol{U}\right)\right)d\tau}_{\textit{Basset history force}}\\ &= F_{SD}+F_{PG}+F_{AM}+F_{BH}.\end{aligned} \tag{10}$$

Here $\boldsymbol{U}$ denotes the undisturbed fluid velocity evaluated at the particle position, $\boldsymbol{U}_p$ is the particle velocity, and D/Dt denotes the material derivative. Faxén corrections proportional to $\nabla^2\boldsymbol{U}$ are included in the Stokes drag, added mass and Basset history terms, accounting for finite-size effects up to $d_p/\eta \approx 4$ (Homann & Bec, 2010), where η is the Kolmogorov length scale. The added mass contribution follows the formulation of Auton *et al.* (1988), which incorporates the effects of local vorticity and rotational motion and is therefore appropriate for strongly sheared and rotational flows.
The relevant control parameter for inertial-particle dynamics is the Stokes number $St = \tau_p/\tau_\eta$, where $\tau_p = \rho_p d_p^2/(18\nu\rho_f)$ is the particle response time and $\tau_\eta = \eta^2/\nu$ is the Kolmogorov time scale.

In this work, we use the second-order, memory-efficient algorithm developed by Van Hinsberg *et al.* (2011), the details of which are briefly described in appendix A for completeness. In addition, appendix B presents a two-step validation of the particle module.
In HST, the particle velocity can be naturally decomposed into a contribution associated with the mean shear and a fluctuating component, in analogy with the fluid velocity decomposition:

$$\boldsymbol{U}_p(t) = \boldsymbol{U}_0\big(\boldsymbol{x}_p(t)\big) + \boldsymbol{u}_p(t),$$

where $\boldsymbol{U}_0\big(\boldsymbol{x}_p(t)\big) = Sz_p(t)$ is the local mean shear evaluated at the particle position $\boldsymbol{x}_p(t)$ and $\boldsymbol{u}_p$ represents the fluctuating velocity.
This decomposition makes it possible to isolate the role of turbulent fluctuations in driving clustering and preferential sampling from the deterministic contribution of the mean flow, which is particularly important in strongly anisotropic configurations such as HST.
In appendix C, the complete derivation of the MRG equation specialized to HST is reported. The resulting formulation shows that the evolution equation for the particle momentum can be expressed entirely in terms of the fluctuating velocity component, with an additional coupling term arising from the mean shear.

2.3.1 Particle interpolation and shear-periodic boundary treatment

The numerical integration of the MRG equation requires the undisturbed fluid velocity, its gradient, and its Laplacian at each particle location. Although spectral interpolation is formally exact (Balachandar & Maxey, 1989), its cost is prohibitive for the large particle numbers considered here. We therefore use a sixth-order three-dimensional Lagrangian interpolation on the physical grid, which offers a good balance between accuracy and efficiency.
In the homogeneous directions (x and y), periodic index wrapping is applied in the standard way. In the vertical direction (z), however, the shear-periodic boundary condition introduces a time-dependent phase shift between the upper and lower boundaries. As a result, direct interpolation across the boundary would be inconsistent unless this shift is taken into account.
To maintain stencil coherence, the shear-periodic condition is enforced in spectral space following equation (4) at the end of each Runge–Kutta substep. After transforming back to physical space, the phase-corrected boundary values are stored in a dedicated array. The interpolation stencil therefore always operates on locally orthogonal data, and no special treatment is needed during interpolation. This ensures that the interpolated velocity field is continuous across the shear-periodic interfaces.
For parallel efficiency, all velocity-related fields required by the particle solver (velocity, gradient, and Laplacian) are stored in MPI (Message Passing Interface) panels aligned with the domain decomposition. Each processor has local access to the field values needed for particles in its subdomain, minimizing inter-processor communication during the Lagrangian update.
Particle crossings of the shear-periodic boundaries are handled using the Lees–Edwards remapping procedure (Lees & Edwards, 1972). For a domain of size (L_x, L_y, L_z) with mean shear rate S, a particle leaving the domain through the upper or lower z boundary is remapped as

$$(x_p, y_p, z_p) \;\mapsto\; \big(x_p \pm StL_z + lL_x,\, y_p + mL_y,\, z_p + nL_z\big), \tag{11}$$

where l, m, and n enforce periodicity in x, y, and z, and the shift $\pm StL_z$ accounts for the relative displacement of the sheared images. To remain consistent with the background mean profile $U_0(z) = Sz$, the particle streamwise velocity is corrected upon remapping:

$$U_p \;\mapsto\; U_p \pm SL_z. \tag{12}$$

This preserves continuity of the particle velocity relative to the local mean flow across the shear-periodic interface. Combined with the phase-consistent interpolation described above, this approach provides a fully consistent treatment of particle advection and force evaluation in homogeneous shear flow.

2.4. Computational details

HST is fully characterized by three nondimensional parameters: the two aspect ratios of the computational domain, $A_{xy} = L_x/L_y$ and $A_{zy} = L_z/L_y$, and the shear Reynolds number $Re_y = SL_y/\nu$. The present simulation corresponds to the smallest case (L32) reported by Sekimoto *et al.* (2016), where "L" denotes the low-Reynolds-number regime and the digits 3 and 2 indicate the values of A_{xy} and A_{zy}, respectively.
The simulation is run until $ST_{\mathrm{stat}} = 2032$, ensuring fully developed statistically stationary conditions. In addition to Re_y, the flow is characterized by the Taylor-scale Reynolds number

$$Re_\lambda = \sqrt{\frac{5}{3\nu\langle\epsilon\rangle}}\,\langle u_i u_i\rangle, \tag{13}$$

and by the Corrsin shear parameter

$$S^* = \frac{S\langle u_i u_i \rangle}{\langle \epsilon \rangle}, \tag{14}$$

where $\langle \epsilon \rangle$ denotes the average dissipation rate of turbulent kinetic energy per unit mass, $\langle u_i u_i \rangle$ is twice the turbulent kinetic energy per unit mass, and repeated indices imply summation.

As described in section 2.2.1, the domain is discretized using $N_x \times N_y \times N_z$ Fourier modes, corresponding to grid resolutions $\Delta x/\eta \approx 0.7$, $\Delta y/\eta \approx 0.5$, and $\Delta z/\eta \approx 1.0$, where η is the Kolmogorov length scale. This resolution is sufficient to fully capture the dissipative range and is consistent with previous DNS studies of HST (Rosti *et al.*, 2019; Yousefi *et al.*, 2020).

Starting from the statistically steady velocity field, inertial particles are introduced in six distinct populations of $N_p = 5 \times 10^5$ particles each. Particles are initialized with random, spatially homogeneous positions and with velocities equal to the local fluid velocity. Three Stokes numbers are considered, $St = [0.1, 0.5, 1]$, obtained by varying the particle-to-fluid density ratio and the particle diameter, as reported in table 2.

The particle equations of motion are integrated using a fixed time step $\Delta t = 3.3 \times 10^{-3}$, which satisfies the stability and accuracy constraints discussed in section 2.2.3. The particle velocity is advanced in time using the semi-implicit scheme proposed by Van Hinsberg *et al.* (2011), while particle positions are updated using a second-order Adams–Bashforth method.

The Basset history term is evaluated using the window-tail decomposition, as described in appendix A. For the explicit window contribution, five previous time steps are retained, following the recommendations of Van Hinsberg *et al.* (2011) and the implementation of Olivieri *et al.* (2014). For the exponential tail approximation, whose sensitivity analysis is reported in appendix D, the coefficient set of Van Hinsberg *et al.* (2011) with $m = 10$ exponentials is adopted, in order to remain consistent with previous studies and to ensure an optimal compromise between accuracy and computational cost.

After an initial transient of $50\,S^{-1}$, statistics are collected in order to ensure independence from the arbitrary initial particle distribution. Particle clustering statistics are computed from 120 independent snapshots separated by S^{-1}, while preferential sampling statistics are accumulated every $5\Delta t$ to provide finer temporal resolution of the sampled flow fields.

The parameters of the carrier flow and of the six particle populations are reported in tables 1 and 2, respectively. A three-dimensional visualization of the flow field and particle distribution in HST is shown in figure 1.

Case	$N_x \times N_y \times N_z$	Re_y	Re_λ	S^*	η	L_s/η
L32	$190 \times 94 \times 192$	2000	47.6 ± 6.4	6.92 ± 0.84	0.0110 ± 0.0026	11

Table 1: Flow parameters for the L32 HST simulation. All reported values are computed by averaging over the statistically stationary interval $ST_{\text{stat}} = 2032$–2202.

Case	St	ρ_p/ρ_f	d_p/η	Re_p	N_p
P1	1.054	10	1.377	2.05 ± 1.40	5×10^5
P2	1.054	100	0.435	0.724 ± 0.456	5×10^5
P3	0.527	10	0.974	0.896 ± 0.664	5×10^5
P4	0.527	100	0.308	0.312 ± 0.209	5×10^5
P5	0.105	10	0.435	0.105 ± 0.130	5×10^5
P6	0.105	100	0.138	0.035 ± 0.026	5×10^5

Table 2: Particle parameters for the six populations considered in the L32 simulation.

3. Homogeneous shear turbulence statistics

In this section we analyze the carrier-flow statistics of the L32 simulation, focusing on scale separation and small-scale structure. These properties are central to the present study, as they directly govern inertial-particle dynamics, preferential sampling, and clustering.

3.1. Energy spectrum

We first examine the one-dimensional energy spectrum of the carrier flow. In statistically stationary HST, the turbulent kinetic energy is continuously produced by the mean shear. Unlike HIT, where the largest scales are

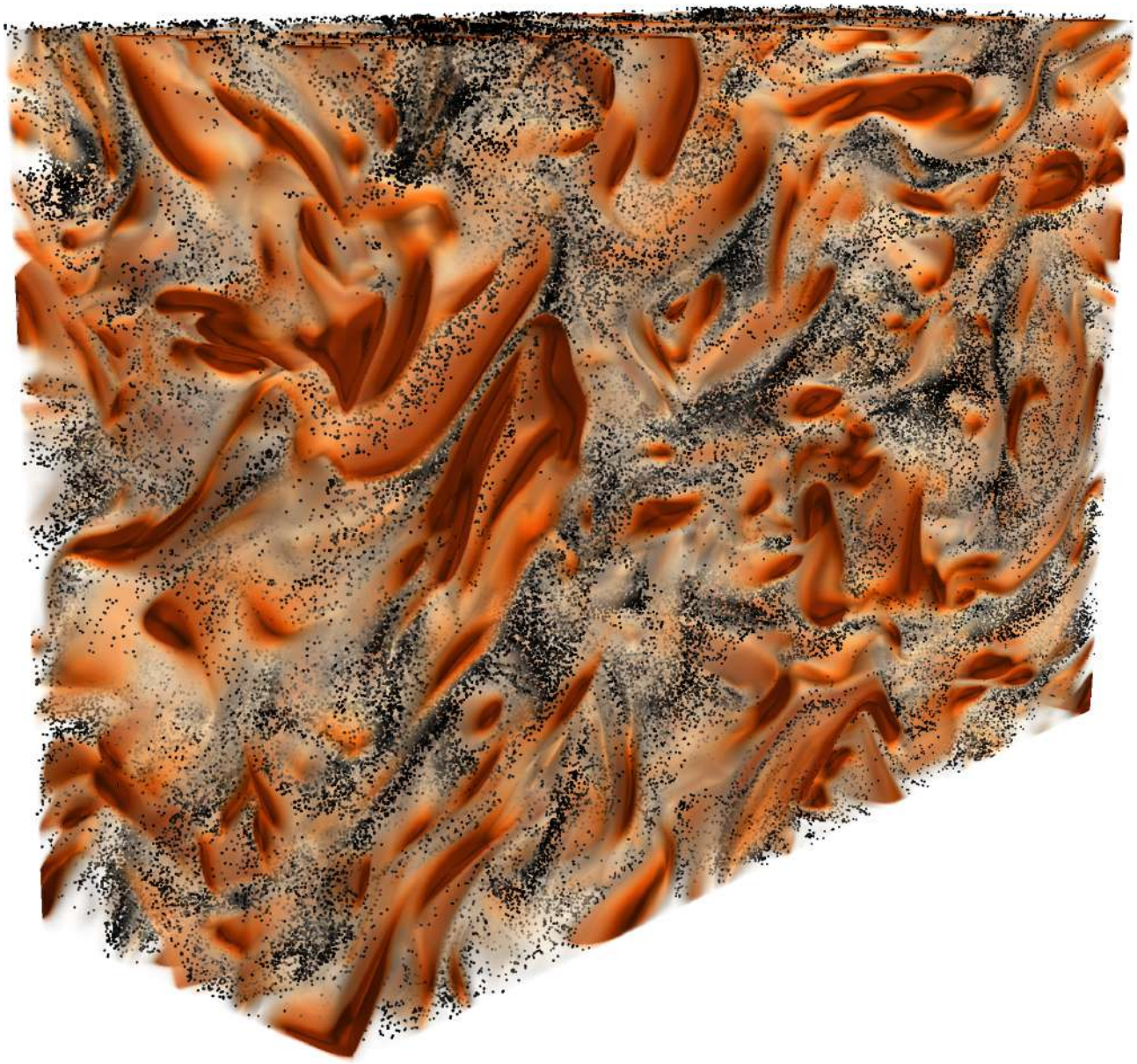


Figure 1: Volumetric rendering of a flow snapshot for the case with $\rho_p/\rho_f = 10$ and $St = 1$. Darker regions indicate higher enstrophy ω^2. The black markers denote particle positions, showing their tendency to preferentially sample low-enstrophy regions of the flow (see section 7.1).

set by the forcing mechanism or by the domain size, in HST the finite computational box directly constrains the largest energetic motions.

In particular, the spanwise dimension L_y largely determines the size of the dominant turbulent structures. As shown by Sekimoto *et al.* (2016), the box aspect ratios $A_{xy} = L_x/L_y$ and $A_{zy} = L_z/L_y$ play a key role in minimizing artificial confinement effects. The L32 configuration used here corresponds to the smallest domain presented by Sekimoto *et al.* (2016) that still sustains statistically meaningful turbulence dynamics.

Figure 2 shows that a large fraction of the kinetic energy resides at the largest scales, which are directly energized by the mean velocity gradient. These motions are strongly anisotropic and dominate turbulent kinetic energy production.

A key length scale in HST is the Corrsin shear scale,

$$L_s = \left(\frac{\langle \epsilon \rangle}{S^3}\right)^{1/2}, \tag{15}$$

which separates scales directly affected by the mean shear from those dominated by nonlinear inertial interactions (Corrsin, 1958). The onset of Kolmogorov scaling is marked in figure 2 using the empirical criterion $kL_s = 0.6$ proposed by Shen & Warhaft (2000). At scales $r \ll L_s$, turbulent eddies progressively decouple from the mean shear and the flow approaches local isotropy.

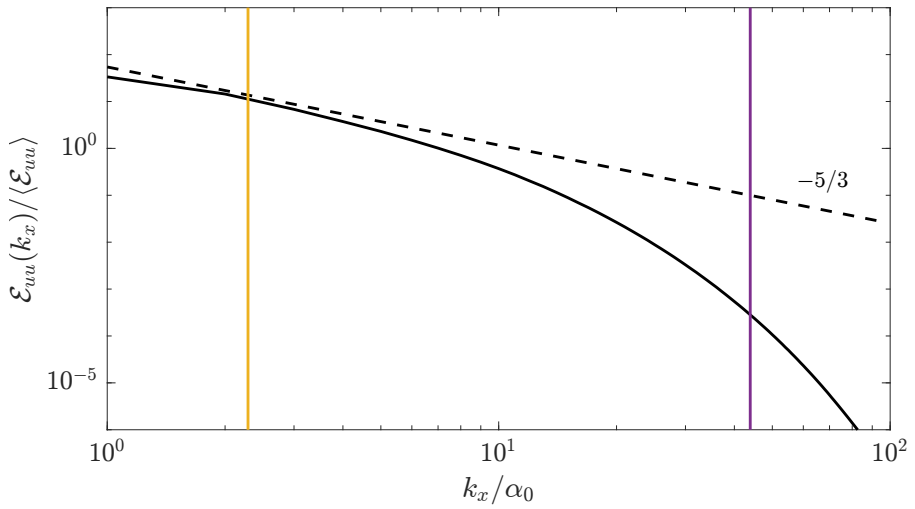


Figure 2: One-dimensional energy spectrum of the streamwise velocity component $\mathcal{E}_{uu}(k_x)$, normalized by its mean, for the L32 HST case. The streamwise wavenumber is normalized by the fundamental mode $\alpha_0 = 2\pi/L_x$. The dashed line shows the Kolmogorov $k^{-5/3}$ scaling. The vertical yellow marker indicates the onset of the inertial range according to the criterion $kL_s = 0.6$ (Shen & Warhaft, 2000), based on the Corrsin shear scale. The purple marker denotes the Kolmogorov wavenumber $k\eta = 2\pi/\eta$.

Following Gualtieri *et al.* (2002), the inertial range in HST can be divided into two subranges. At intermediate scales ($L \gg r \gg L_s$), the dynamics remains shear dominated and energy production is significant. At smaller scales ($L_s \gg r \gg \eta$), shear effects become negligible and the flow follows the classical inertial cascade.
The relative importance of shear is quantified by the Corrsin shear parameter

$$S^* = \frac{S\langle u_i u_i\rangle}{\langle\epsilon\rangle} = \left(\frac{l_d}{L_s}\right)^{2/3}, \tag{16}$$

where $l_d = (2\langle u_i u_i\rangle)^{3/2}/\langle\epsilon\rangle$ is an energy-containing length scale. Large values of S^* indicate that a substantial fraction of the dynamically active scales is influenced by shear. The separation between dissipative and shear-dominated ranges is measured by

$$S_c^* = \left(\frac{\eta}{L_s}\right)^{2/3}, \tag{17}$$

which plays a role analogous to the Taylor-scale Reynolds number in HIT.
For the L32 case, $S^* \approx 6.92$ and $S_c^* \approx 0.2$, in agreement with the values reported in Gualtieri *et al.* (2002). These values indicate a clear hierarchy of scales, with a pronounced shear-dominated range coexisting with a classical inertial subrange. Nevertheless, due to the moderate Reynolds number and the persistent influence of shear, the inertial range remains short, spanning less than one decade in wavenumber.

3.2. Local structure of the flow

We now examine the small-scale structure of the carrier flow, which is particularly relevant for inertial-particle dynamics and clustering, as these processes are governed by the local velocity gradients sampled by the particles. At sufficiently small scales, the velocity field is smooth and can be approximated locally by a first-order Taylor expansion around a point $\boldsymbol{x}_0$,

$$u_i(\boldsymbol{x}, t) = u_i(\boldsymbol{x}_0, t) + A_{ij}(\boldsymbol{x}_0, t)(x_j - x_{0,j}) + \mathcal{O}(|\boldsymbol{x} - \boldsymbol{x}_0|^2), \tag{18}$$

where $A_{ij} = \partial u_i/\partial x_j$ is the velocity gradient tensor. This approximation is valid over distances of order the Kolmogorov scale η, over which spatial variations of A_{ij} are weak (Meneveau, 2011). As a result, the local flow topology at dissipative scales is fully characterized by A_{ij}.
The velocity gradient tensor is decomposed into symmetric and antisymmetric parts,

$$S_{ij} = \frac{A_{ij} + A_{ji}}{2}, \qquad W_{ij} = \frac{A_{ij} - A_{ji}}{2}, \tag{19}$$

where S_{ij} is the strain-rate tensor and W_{ij} the rotation-rate tensor. The local kinematics is commonly described by (Ashurst *et al.*, 1987; Meneveau, 2011; Davidson, 2015): *(i)* the eigenvalues $\alpha \geq \beta \geq \gamma$ of S_{ij}, *(ii)* the vorticity

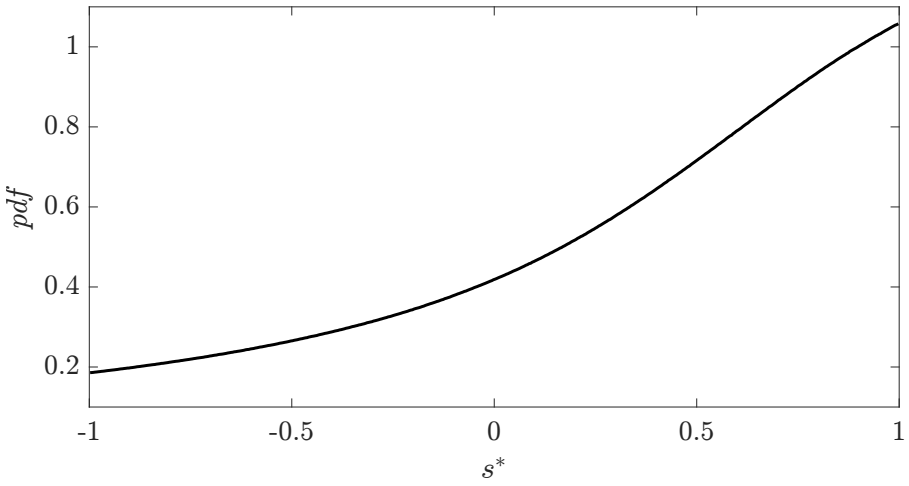


Figure 3: Probability density function of the strain-state parameter s^* for the L32 case.

vector $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ and its magnitude ω^2, and *(iii)* the alignment between $\boldsymbol{\omega}$ and the strain eigenvectors. For incompressible flow, $\alpha + \beta + \gamma = 0$, implying that $\alpha \geq 0$, $\gamma \leq 0$, while β can take either sign.
To quantify the strain geometry we use the strain-state parameter introduced by Lund & Rogers (1994),

$$s^* = \frac{3\sqrt{6}\,\alpha\beta\gamma}{(\alpha^2 + \beta^2 + \gamma^2)^{3/2}}. \tag{20}$$

In a random velocity-gradient field s^* is uniformly distributed, whereas in turbulence it reflects preferential strain topologies.
As shown in figure 3, the distribution of s^* exhibits a peak near $s^* = 1$, in agreement with the well-established result that axisymmetric extension is the most likely straining state in purely Newtonian turbulence (Meneveau, 2011; Davidson, 2015). This disk-like configuration is the most probable state in three-dimensional turbulence and may arise from both vortex sheet formation and the stretching of vortex tubes (Davidson, 2015).
The probability density function of the enstrophy ω^2 and the alignment between vorticity and strain eigenvectors are shown in figure 4. The enstrophy distribution exhibits a pronounced long tail, indicating strong intermittency and the presence of intense, localized vortex filaments.

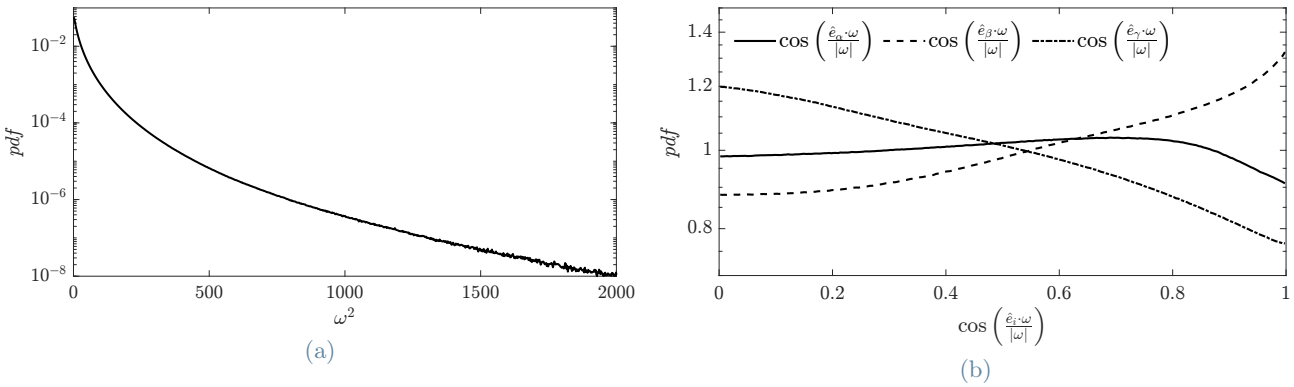


Figure 4: (a) Probability density function of the enstrophy ω^2. (b) Alignment of the vorticity vector $\boldsymbol{\omega}$ with the principal strain eigenvectors $\hat{\boldsymbol{e}}_\alpha$, $\hat{\boldsymbol{e}}_\beta$ and $\hat{\boldsymbol{e}}_\gamma$.

Consistent with HIT, vorticity preferentially aligns with the intermediate strain eigenvector $\hat{\boldsymbol{e}}_\beta$ and tends to be orthogonal to the most compressive direction $\hat{\boldsymbol{e}}_\gamma$.
This reflects vortex stretching, whereby vorticity is amplified while avoiding direct compression (Ashurst *et al.*, 1987; Davidson, 2015). The persistence of this behavior in HST indicates that, despite strong mean shear, the small-scale dynamics retains many universal features of three-dimensional turbulence, in agreement with the scale separation discussed in section 3.1.
Further insight is obtained from the invariants of A_{ij}. The eigenvalues λ satisfy

$$\lambda^3 + P\lambda^2 + Q\lambda + R = 0, \tag{21}$$

with invariants (Chong *et al.*, 1990; Cantwell, 1993; Meneveau, 2011)

$$P = \mathrm{tr}(A_{ij}) = 0, \qquad Q = \frac{\omega^2}{4} - \frac{1}{2}(\alpha^2 + \beta^2 + \gamma^2), \qquad R = -\frac{1}{3}(\alpha^3 + \beta^3 + \gamma^3) - \frac{1}{4}\,\omega_i \omega_j S_{ij}. \tag{22}$$

The discriminant of the characteristic polynomial,

$$\Delta = \frac{27}{4} R^2 + Q^3, \tag{23}$$

distinguishes regions with complex conjugate eigenvalues ($\Delta < 0$, vortical topology) from those with three real eigenvalues ($\Delta > 0$, strain-dominated topology).

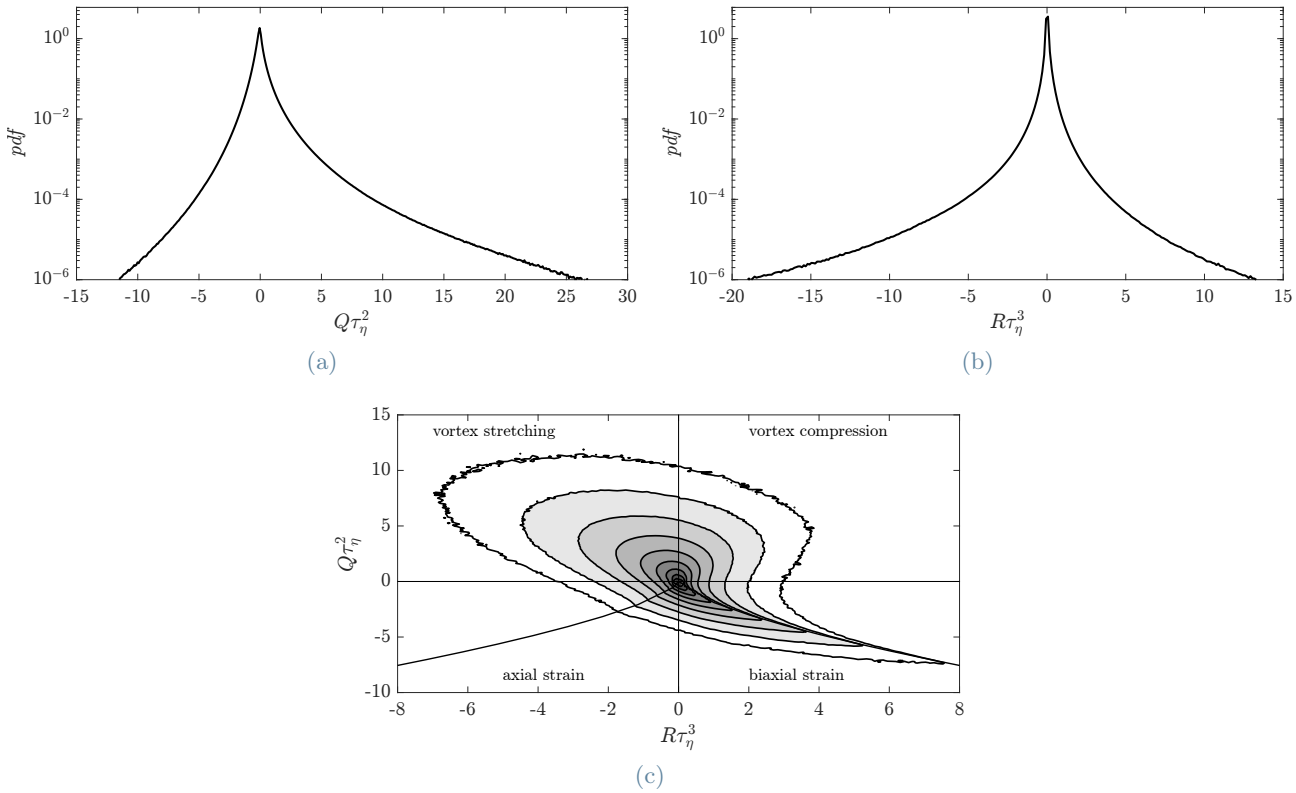


Figure 5: Top: marginal distributions of Q (a) and R (b). Bottom: joint probability density of Q and R with 10 contour levels between 5×10^{-6} and 10. The thin black curves denote the Vieillefosse tails ($\Delta = 0$).

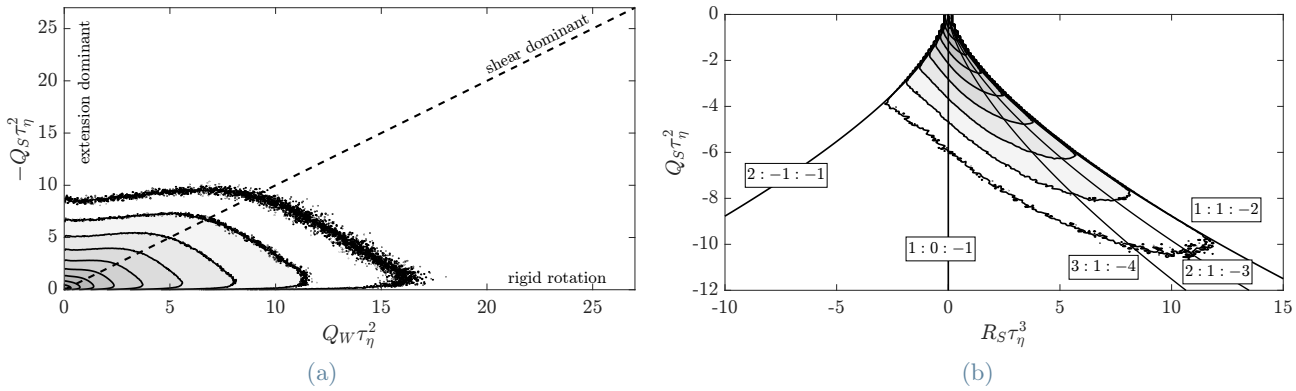


Figure 6: (a) Joint probability density of Q_S and Q_W. (b) Joint probability density of Q_S and R_S, with labels indicating strain eigenvalue ratios. Both panels use 10 contour levels between 5×10^{-6} and 10.

The joint distribution of Q and R (figure 5) displays the characteristic tear-drop shape observed in turbulent flows. The probability density extends preferentially along the right Vieillefosse tail ($Q < 0$, $R > 0$), indicating the prevalence of strong strain-dominated events associated with biaxial stretching. The highest probability occurs in the second and fourth quadrants, consistent with the expected negative correlation between Q and R in three-dimensional turbulence (Davidson, 2015). The concentration of probability near the origin and the near-zero mean values of Q and R confirm the statistical homogeneity of the flow (Nomura & Post, 1998).
Additional insight is provided by considering the invariants of the strain-rate and rotation-rate tensors separately,

$$Q_S = -\frac{1}{2}(\alpha^2 + \beta^2 + \gamma^2), \qquad Q_W = \frac{1}{4}\,\omega^2, \qquad R_S = -\frac{1}{3}(\alpha^3 + \beta^3 + \gamma^3). \tag{24}$$

The joint distribution of Q_S and Q_W (figure 6a) shows that events with $Q_W > -Q_S$ are more frequent, indicating that rotation-dominated regions are statistically more prevalent than purely extensional ones. The joint distribution of Q_S and R_S (figure 6b) is skewed toward positive R_S, demonstrating that biaxial strain states occur more frequently than axial strain states. This reflects the predominance of sheet-like strain structures over tube-like configurations, a robust feature of three-dimensional turbulence (Chong *et al.*, 1990; Ooi *et al.*, 1999).

4. Force balance in the Maxey–Riley–Gatignol equation

In this section we examine the relative importance of the force contributions in the MRG equation and their role in shaping inertial-particle dynamics in HST. Similar analyses have been performed for HIT by Olivieri *et al.* (2014) and by Daitche (2015), leading to markedly different conclusions regarding the relevance of the Basset history force.
As discussed by Daitche (2015), this discrepancy mainly arises from differences in the forcing strategy of the carrier flow. In Daitche's simulations, turbulence is forced on long time scales, whereas in Olivieri's simulations both the fluid and the particles are forced on comparatively short time scales. This distinction affects the temporal correlations of the particle slip velocity and, consequently, the magnitude of the history contribution. The present HST configuration is unforced and sustained solely by mean shear production. It therefore provides a complementary framework for assessing which of these scenarios more accurately represents the particle-force balance in shear-driven turbulence.

4.1. Relative magnitude of the Basset history force

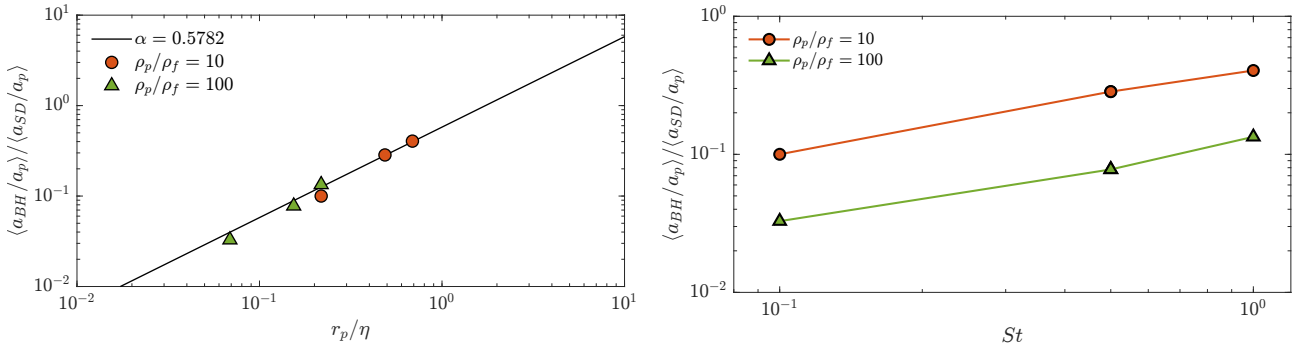
From the MRG model (equation (10)), the particle acceleration $\boldsymbol{a}_p$ can be written as

$$\boldsymbol{a}_p = \boldsymbol{a}_{SD} + \boldsymbol{a}_{PG} + \boldsymbol{a}_{AM} + \boldsymbol{a}_{BH}, \tag{25}$$

where the terms denote contributions from Stokes drag, pressure gradient, added mass, and the Basset history force, respectively.
A useful way to assess the importance of the Basset history term and to evaluate the validity of the simplified models is to consider the ratio of the Basset history term mean magnitude to the Stokes drag. Using the fractional time-derivative structure of the Basset term, Daitche (2015) derived a simple scaling

$$\frac{a_{BH}}{a_{SD}} \approx \alpha\sqrt{\frac{3\beta\tau_p}{\tau_\eta}} = \alpha\frac{r_p}{\eta}, \tag{26}$$

where r_p is the particle radius, η is the Kolmogorov scale, $\beta = 3/(2\rho_p/\rho_f + 1)$ accounts for density ratio, and α is an empirical prefactor. This relation emphasizes that the Basset history force grows with particle size: small particles respond quickly to the flow and "forget" past accelerations, while larger particles retain memory, enhancing the Basset contribution.
Figure 7a shows the ratio $\langle a_{BH}/a_p\rangle/\langle a_{SD}/a_p\rangle$ as a function of r_p/η. The trend follows the predicted scaling, with only a weak dependence on particle density. We report this ratio rather than the one in equation (26), as it provides a more consistent estimate of all force contributions discussed in section 4.2. A least-squares fit gives $\alpha \approx 0.58$, slightly smaller than the HIT value $\alpha \approx 0.69$ reported by Daitche (2015), likely reflecting differences in flow configuration, dataset size, and the non-identical force ratio. Overall, the scaling remains robust.
Figure 7b shows the same ratio as a function of the Stokes number. At fixed Stokes number, increasing the density ratio reduces the relative importance of the Basset history force. At fixed density ratio, decreasing St also weakens the history contribution. This behavior reflects that St combines both particle size and density;



(a) (b)

Figure 7: Relative magnitude of the Basset history force compared to Stokes drag, expressed as the ratio $\langle a_{BH}/a_p\rangle/\langle a_{SD}/a_p\rangle$. Results are plotted against (a) the normalized particle radius r_p/η and (b) the Stokes number St.

variations at fixed St correspond effectively to changes in size, consistent with the physical picture of Basset forces being more important for larger particles.
From figure 7a, the relevance of the Basset history force in HST can be summarized as follows

- **Small particles ($r_p/\eta \sim 10^{-2}$):** the Basset history force is negligible, of order $\mathcal{O}(1\%)$ of the Stokes drag.
- **Larger particles ($r_p/\eta \sim 1$):** the Basset history force becomes comparable to Stokes drag, significantly influencing particle dynamics.

These conclusions closely mirror those reported by Daitche (2015) for HIT, confirming the scaling of the Basset force with particle size.

4.2. Relative importance of all terms in the MRG equation

We now examine the full force balance among all MRG contributions. Figure 8 shows the probability density functions (PDFs) of the normalized contributions a_i/a_p, which satisfy

$$\frac{a_{SD}}{a_p} + \frac{a_{PG}}{a_p} + \frac{a_{AM}}{a_p} + \frac{a_{BH}}{a_p} = 1.$$

Using the Basset scaling (equation (26)), an estimate of the slip velocity, and neglecting Faxén corrections, Daitche (2015) derived approximate expressions for the normalized accelerations:

$$\frac{a_{SD}}{a_p} \approx \frac{\xi(1-\beta)}{1+\alpha\frac{r_p}{\eta}(1-\beta)}, \qquad \frac{a_{BH}}{a_p} \approx \frac{\xi\alpha\frac{r_p}{\eta}(1-\beta)}{1+\alpha\frac{r_p}{\eta}(1-\beta)} \tag{27}$$

$$\frac{a_{PG}}{a_p} \approx \frac{2}{3}\frac{\xi\beta}{1+\alpha\frac{r_p}{\eta}(1-\beta)}, \qquad \frac{a_{AM}}{a_p} \approx \frac{1}{2}\left(\frac{a_{PG}}{a_p} - \frac{\rho_f}{\rho_p}\right) \tag{28}$$

where $\xi = (2\rho_p/\rho_f + 1)/(2\rho_p/\rho_f)$ and $\beta = 3/(2\rho_p/\rho_f + 1)$.
From figure 8, it is clear that the approximations capture the mean force balance well when the Basset force is small. As the history term grows, particularly for larger particles (figures 8a and 8c), the model slightly underestimates the pressure gradient contribution and overestimates Stokes drag, reflecting the fact that part of the acceleration attributed to instantaneous drag is actually mediated by memory effects. Despite this, the mean magnitude of the Basset force is consistently well predicted, confirming that its scaling with particle size accurately reflects the physical influence of past accelerations. Increasing particle density suppresses both pressure gradient and history effects, leaving Stokes drag as the dominant contribution, while the distribution tails reveal intermittent events where memory and pressure forces transiently become significant. Overall, figure 8 reinforces the physical picture that the Basset history force matters primarily for larger particles, acting as a memory term that redistributes instantaneous acceleration among the MRG contributions without altering the overall qualitative trends.
In summary: *(i)* the scaling arguments introduced by Daitche (2015) for HIT remain valid in HST; *(ii)* the relative importance of the Basset history force depends primarily on particle size and only weakly on density; *(iii)* for reduced models focused on mean dynamics, Daitche's estimates reliably approximate the average contributions of all MRG terms; and *(iv)* the non-negligible role of the Basset history force for large particles, highlighted by Olivieri *et al.* (2014), is confirmed, though their study overestimates its quantitative magnitude.

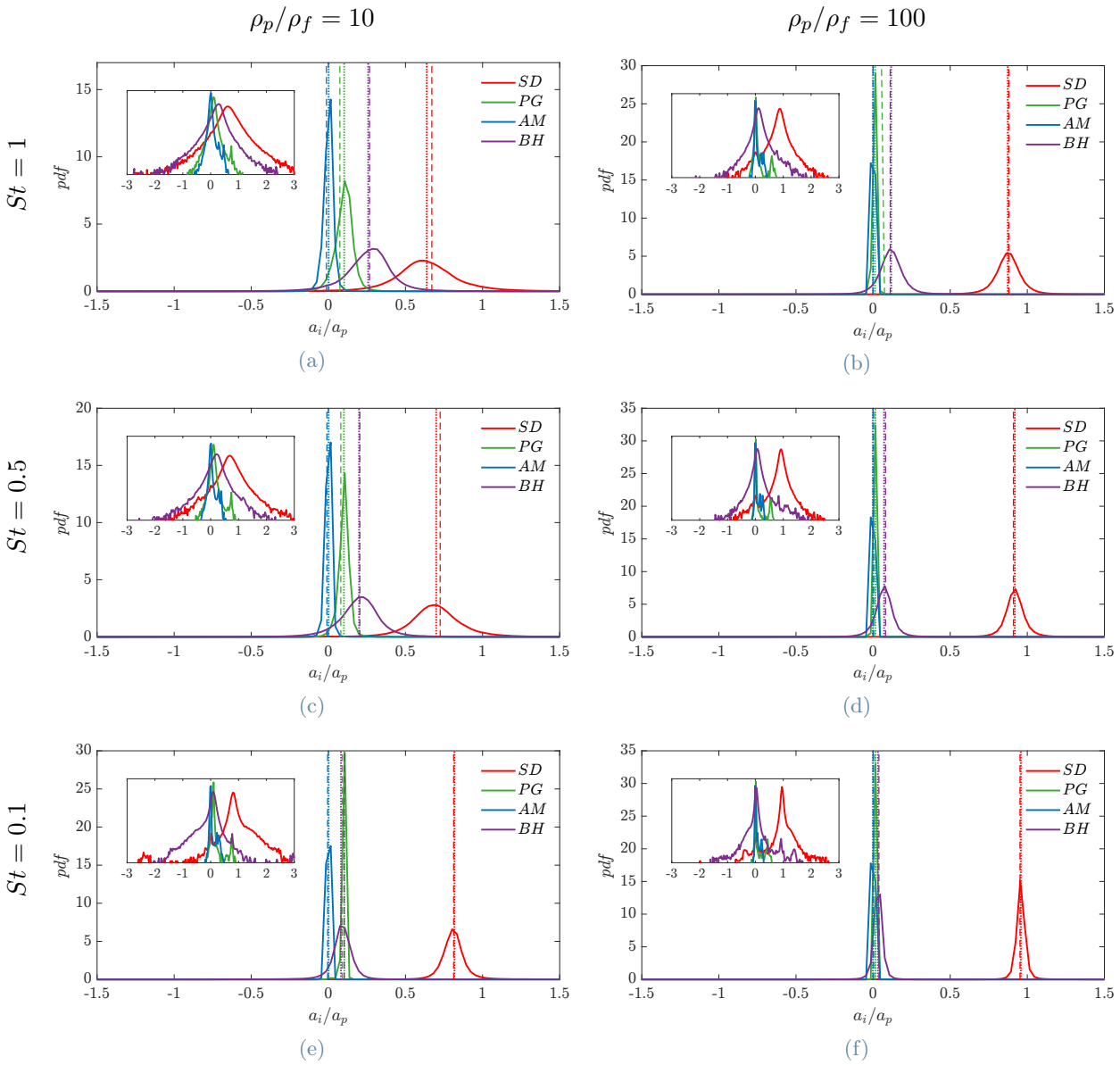


Figure 8: Probability density functions of the normalized force contributions a_i/a_p. Main panels use linear–linear scales, while insets use linear–logarithmic scales to emphasize distribution tails. Rows correspond to different Stokes numbers ($St = 1$, 0.5, 0.1 from top to bottom), and columns to different density ratios ($\rho_p/\rho_f = 10$ and 100). Vertical dotted lines show mean values from the simulations; dashed lines indicate the theoretical estimates of Daitche (2015).

5. Clustering, anisotropy, and spatial organization of inertial particles

Figure 9 shows instantaneous particle distributions for $\rho_p/\rho_f = 10$ at two Stokes numbers ($St = 1$ and $St = 0.1$). Particles are overlaid on the magnitude of the vorticity component normal to the slice. The particle field is strongly inhomogeneous, with voids correlated with intense vorticity regions and thin filamentary clusters where particles preferentially accumulate. These features are the classical signature of inertial clustering in turbulence and are closely linked to high-enstrophy regions (Squires & Eaton, 1991; Eaton & Fessler, 1994). The effect of inertia is evident when comparing the two cases. For $St = 0.1$, particles behave as weakly inertial tracers and remain nearly uniformly distributed, with only mild clustering. For $St = 1$, the distribution becomes highly intermittent, with strong small-scale clustering and pronounced voids.

Mean shear introduces an additional anisotropy. As reported by Gualtieri *et al.* (2009), clusters in HST tend to align with the direction of mean stretching. This preferential bottom-left to top-right orientation is clearly visible here, particularly in the x–z plane for $St = 1$, indicating that the mean velocity gradient influences particle organization even at dissipative scales.

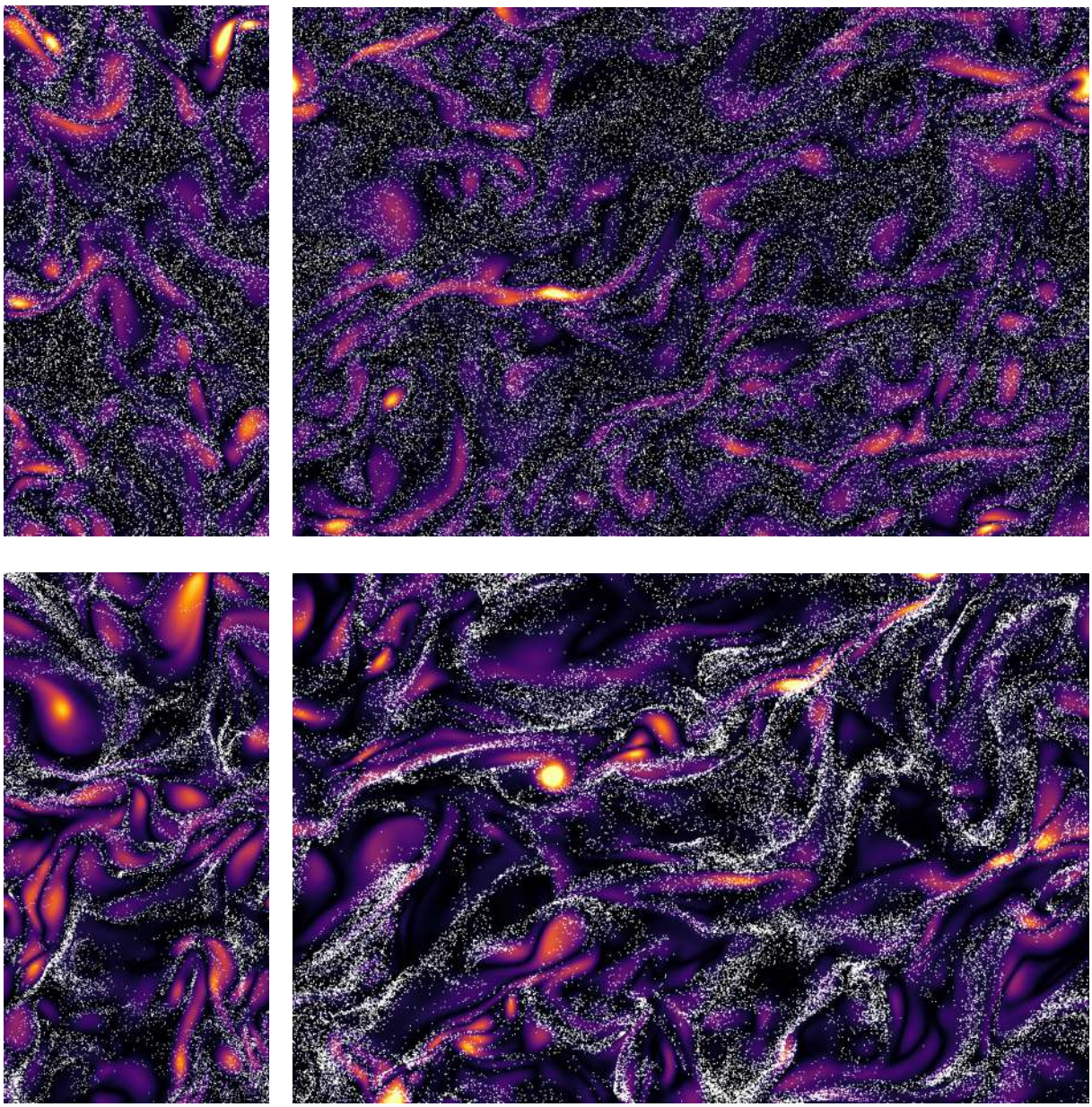


Figure 9: Instantaneous particle configurations for $\rho_p/\rho_f = 10$ (white dots), superimposed on the modulus of the vorticity component normal to the plane (lighter regions correspond to higher vorticity). From top to bottom: $St = 0.1$ and $St = 1$. Left column: thin slice in the y–z plane; right column: slice in the x–z plane. The slice thickness is approximately 10 Kolmogorov scales.

5.1. Radial distribution function

Small-scale clustering of particles can be quantified using the radial distribution function (RDF) $g(r)$ (Sundaram & Collins, 1997), defined as

$$g(r) = \frac{N_s(r)/\Delta V(r)}{N_{pa}/V}, \tag{29}$$

where $N_s(r)$ is the number of particle pairs within a shell of thickness $2\Delta r$ at separation r, $\Delta V(r)$ is the shell volume, $N_{pa} = N_p(N_p-1)/2$ is the total number of particle pairs, and V is the domain volume. For a spatially uniform particle distribution, $g(r) = 1$. Values $g(r) > 1$ indicate clustering, while $g(r) < 1$ indicates depletion at scale r.

Figure 10 confirms the trends already suggested by particle snapshots. For a density ratio $\rho_p/\rho_f = 10$, weakly inertial particles ($St = 0.1$) exhibit $g(r) \approx 1$ over most separations, consistent with a nearly uniform distribution. In contrast, for $St = 1$ the RDF increases sharply as $r \to 0$, revealing intense small-scale clustering. At fixed St, increasing particle-to-fluid density ratio systematically enhances $g(r)$ over the clustering range. The RDF thus provides direct, scale-dependent evidence that memory effects weaken inertial clustering (Olivieri *et al.*,

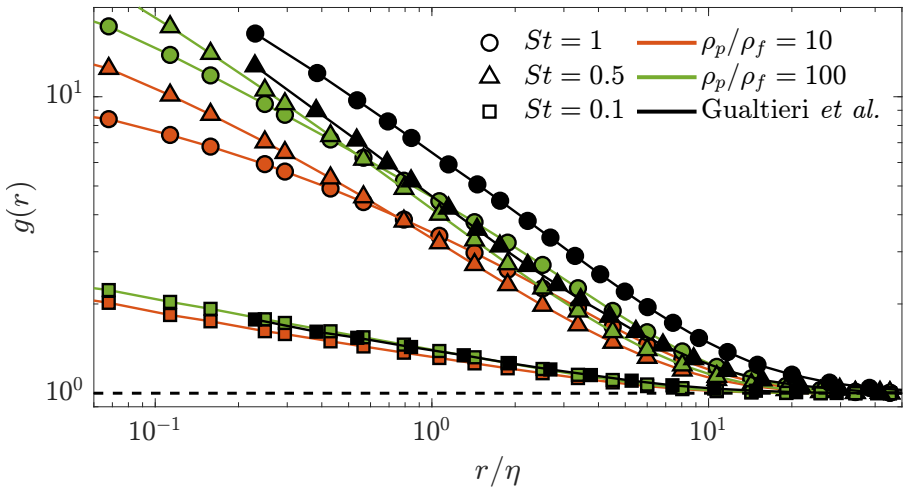


Figure 10: Radial distribution function $g(r)$ for all cases considered. Black markers denote the results of Gualtieri *et al.* (2009), included for comparison with the simplified MRG model.

2014; Daitche, 2015), because, as discussed in section 4.1, for a given St higher density ratios correspond to effectively smaller particles and hence weaker memory effects.
Comparison with the simplified MRG model results of Gualtieri *et al.* (2009) highlights this effect. For $St = 0.1$, the two datasets coincide, confirming that the history force is negligible at low inertia. For larger St, however, differences appear: while drag-only models predict monotonically stronger clustering with increasing St, our full MRG simulations show that the expected ordering between $St = 0.5$ and $St = 1$ holds only down to $r/\eta = \mathcal{O}(1)$. At smaller scales, enhanced memory effects at $St = 1$ reduce the growth of $g(r)$ and lead to earlier saturation.
At sufficiently small separations, the RDF follows a power-law scaling

$$g(r) \propto r^{\mathcal{D}_2 - 3},$$

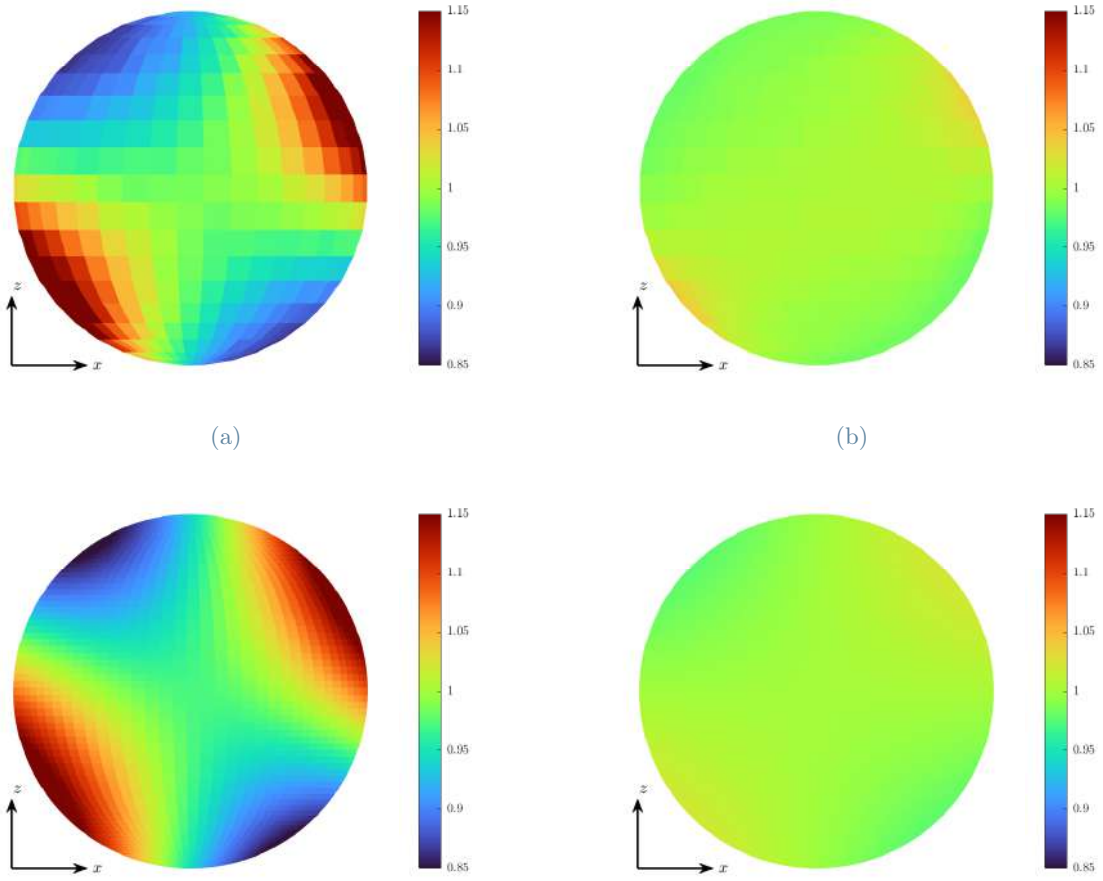
where $\mathcal{D}_2$ is the correlation dimension (Grassberger & Procaccia, 1983; Bec, 2003). Fittings over $r/\eta \in [0.1, 10]$ yield $\mathcal{D}_2 \approx 2.9$ for $St = 0.1$, confirming weak clustering, and $\mathcal{D}_2 \approx 2.4$–$2.5$ for $St = 0.5$ and 1, consistent with previous HIT results (Bec *et al.*, 2007; Daitche, 2015).
For $St = 1$, the power-law scaling breaks down below $r/\eta = \mathcal{O}(1)$, where the RDF saturates and the local correlation dimension approaches the homogeneous limit. This agrees with the theory of Yoshimoto & Goto (2007), who predicted that clustering is active only over a finite range of scales. Specifically, for $St = 1$ the predicted lower cutoff scale $\ell_{\min}/\eta \approx 0.35$ aligns closely with the separation where saturation occurs. Although this scale formally is below the Kolmogorov length, its proximity to the observed cutoff supports the physical relevance of this estimate. For $St = 0.1$, the predicted upper cutoff $\ell_{\max}/\eta \approx 1$ is consistent with the scale at which deviations from homogeneity appear.
In summary, the RDF analysis shows that inertial clustering is limited to a finite range of scales, and that the Basset history force plays a significant role in moderating clustering, particular for larger particles.

5.2. Angular distribution function

While the RDF captures the radial distribution of particle pairs, it contains no information on their relative orientation. In HST, however, anisotropy plays a central role. To quantify it, we use the angular distribution function (ADF) introduced by Gualtieri *et al.* (2009), which generalizes the RDF by retaining full angular information. This diagnostic has subsequently been employed to investigate anisotropic clustering in HST by Nicolai *et al.* (2014) and Lee & Lee (2020).
The ADF is defined as the probability of finding a particle pair within a radial and angular interval:

$$g(r, \theta, \phi) = \frac{N_s(r, \theta, \phi)/\Delta V(r, \theta, \phi)}{N_{pa}/V}, \tag{30}$$

where $N_s(r, \theta, \phi)$ denotes the number of particle pairs in the specified radial and angular intervals and $\Delta V(r, \theta, \phi)$ is the volume of the corresponding spherical sector. Here $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi]$ denote the polar and azimuthal angles, respectively.
Figure 11 illustrates the ADF for ($\rho_p/\rho_f = 10$, $St = 1$). At large separations ($r \approx 35\eta$), the ADF is nearly isotropic. Within the clustering range ($r \approx 4\eta$), a pronounced quadrupolar pattern emerges, with lobes aligned



(a) (b)

(c) (d)

Figure 11: Angular distribution function for $\rho_p/\rho_f = 10$ and $St = 1$. Top row (a-b): full ADF at separations $r \approx 4\eta$ (left) and $r \approx 35\eta$ (right), normalized by the corresponding RDF. Bottom row (c-d): reconstructed ADF retaining only the isotropic ($j = 0$) and quadrupolar ($j = 2$, $m = 0, 1, 2$) sectors.

along the principal directions of the mean strain. This angular organization mirrors the diagonal alignment observed in particle snapshots (figure 9).
Expanding the ADF in spherical harmonics isolates contributions with increasing degrees of anisotropy:

$$g(r, \theta, \phi) = \sum_{j=0}^{\infty} \sum_{m=-j}^{j} g_{jm}(r)\, Y_{jm}(\theta, \phi). \tag{31}$$

The isotropic RDF is recovered as the monopole term:

$$g_{00}(r) \equiv g(r) = \int_{\Omega} g(r, \theta, \phi)\, Y_{00}(\theta, \phi)\, d\Omega. \tag{32}$$

This decomposition projects the ADF onto orthogonal subspaces with increasing degrees of anisotropy: $j = 0$ corresponds to isotropy, $j = 1$ to dipolar patterns, $j = 2$ to quadrupolar anisotropy, and higher j capture progressively finer angular structures. Figure 12a shows a visual representations of the first few real spherical harmonics.
Retaining only the $j = 0$ and $j = 2$ contributions almost fully reconstructs the ADF in figure 11, showing that quadrupolar anisotropy dominates; higher-order modes provide minor corrections, consistent with previous HST studies (Gualtieri *et al.*, 2009).
Figure 12 reports the normalized coefficients $\mathrm{Re}[g_{jm}(r)]/g_{00}(r)$. In all cases, the dominant anisotropic contribution corresponds to $(j, m) = (2, 1)$, a quadrupolar pattern inclined 45° in the shear-vorticity plane. Its persistence across a wide range of separations indicates that anisotropy remains dynamically relevant down to

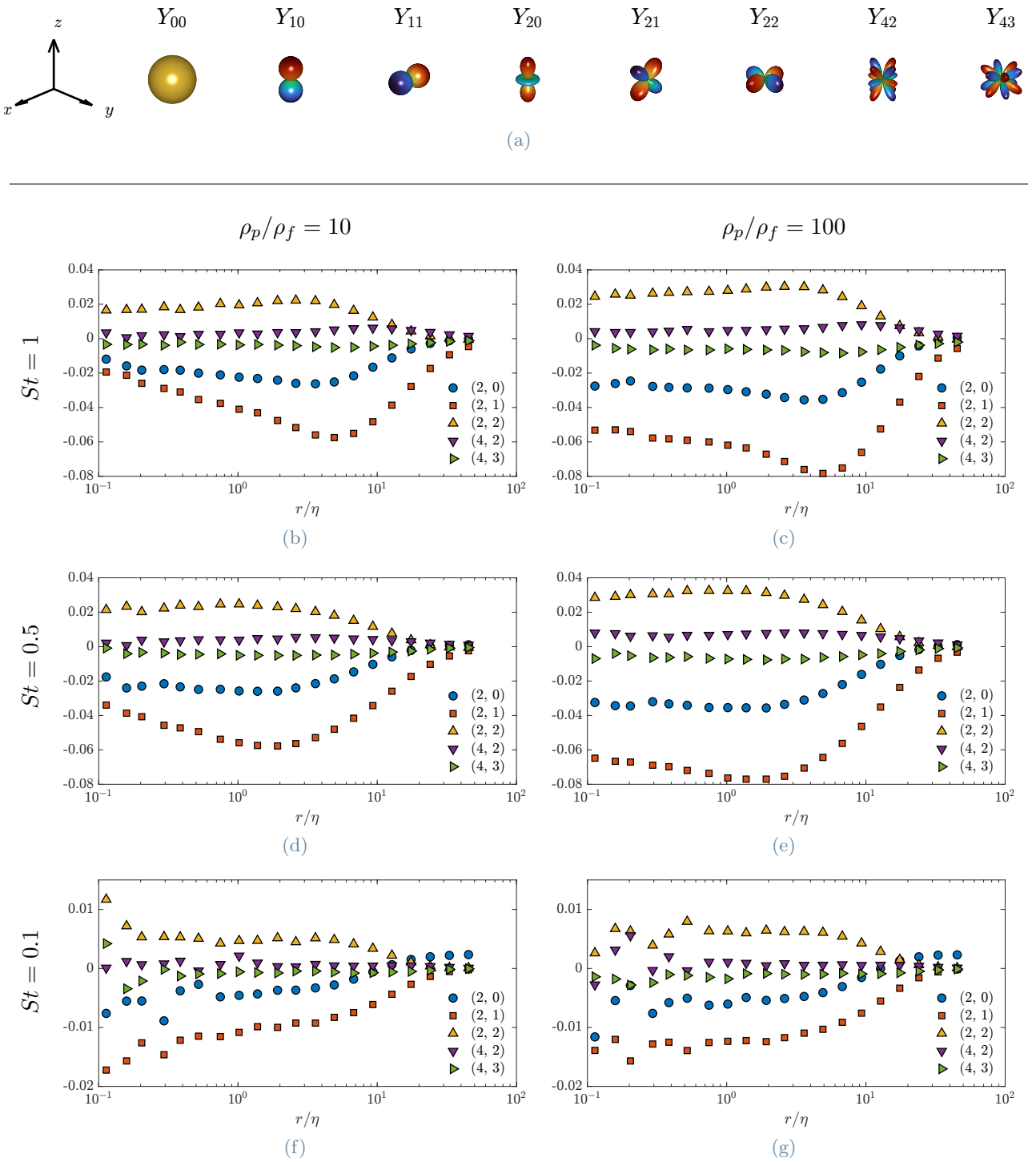


Figure 12: (a) Visual representations of the first few real spherical harmonics. Red and blue regions indicate positive and negative values, respectively, while the radial distance of the surface from the origin represents the absolute magnitude of the harmonic in a given angular direction. (b-g) Normalized amplitudes of selected anisotropic spherical-harmonic sectors, $\mathrm{Re}[g_{jm}(r)]/g_{00}(r)$, as a function of separation. Rows correspond to different Stokes numbers ($St = 1$, 0.5, 0.1 from top to bottom), and columns to different density ratios ($\rho_p/\rho_f = 10$ and 100).

dissipative scales. The negative sign of $\mathrm{Re}[g_{21}(r)]/g_{00}(r)$ reflects the phase of the real part of Y_{21} relative to the observed cluster orientation.

Figure 13 summarizes the scale-dependent intensity of anisotropy. At large separations, all curves collapse toward zero, consistent with isotropy recovery, paralleling the RDF's approach to unity. At smaller scales, $St = 1$ and $St = 0.5$ exhibit pronounced peaks at $r/\eta = \mathcal{O}(10)$, with stronger and more persistent anisotropy for larger density ratios. This trend aligns with the force-balance analysis in section 4.1, which showed that

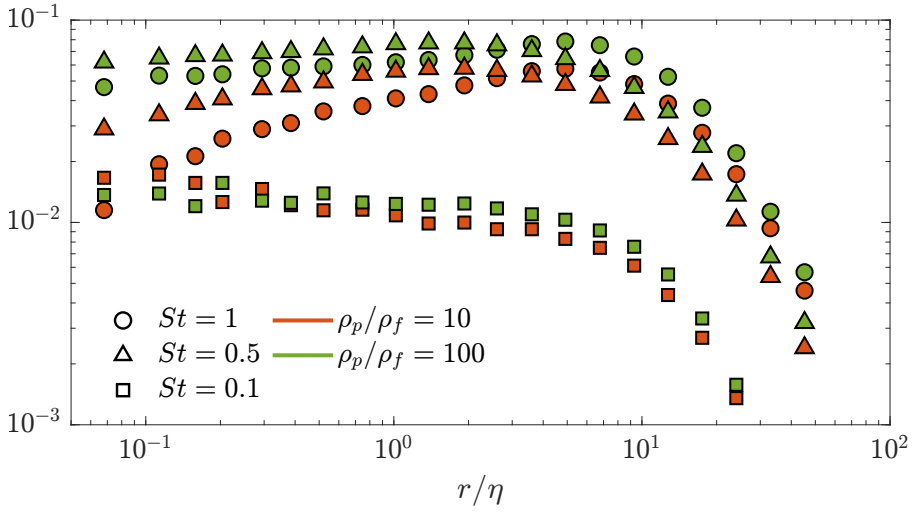


Figure 13: Ratio of the dominant quadrupolar anisotropic contribution $(j, m) = (2, 1)$ to the isotropic RDF $g_{00}(r)$ as a function of separation for all particle sets.

increasing ρ_p/ρ_f at fixed St reduces the relative importance of the Basset history force, enhancing both clustering and anisotropy. Weakly inertial particles ($St = 0.1$) show little scale-dependent amplification, consistent with near-unity RDF and $\mathcal{D}_2 \approx 3$, confirming weak clustering and weak anisotropy. At the smallest separations, the decay of the $(2, 1)$ mode for $\rho_p/\rho_f = 10$ coincides with RDF saturation and the approach to isotropy, whereas this decay is less pronounced at higher density ratios where memory effects are weaker. Overall, anisotropy and clustering are tightly coupled: stronger fractal clustering correlates with stronger and more persistent quadrupolar anisotropy.

5.3. Voronoï tessellation analysis

Particle spatial organization is further quantified using Voronoï tessellation (Monchaux *et al.*, 2010; Monchaux, 2012). Each particle is assigned a cell containing all points in the domain closer to it than to any other particle. The Voronoï cell volume V_V provides a local measure of particle concentration: small volumes correspond to clustered regions, whereas large volumes identify voids.

Voronoï cells are computed using a routine that accounts for shear-periodic boundary conditions, followed by tessellation with the `Voro++` library (Rycroft, 2009). As a reference, we consider a spatially uniform Poisson process, for which the normalized Voronoï volume $\mathcal{V} = V_V/\langle V_V \rangle$ follows the analytical gamma distribution (Ferenc & Néda, 2007)

$$f_3(\mathcal{V}) = \frac{5^5}{\Gamma(5)}\,\mathcal{V}^4\,\exp(-5\mathcal{V}). \tag{33}$$

Figure 14a shows excellent agreement between the analytical expression and a numerical random suspension, validating its use as a reference distribution. As shown in figure 14b, strongly inertial cases ($St = 1, 0.5$) exhibit a pronounced shift toward small $\mathcal{V}$ values and extended tails at large volumes, indicating intense clustering accompanied by large voids. In contrast, weakly inertial cases ($St = 0.1$) remain close to the Poisson reference, in agreement with the RDF and ADF results. At fixed St, increasing the particle-to-fluid density ratio enhances the probability of very small Voronoï volumes, confirming stronger clustering as memory effects weaken. Voronoï statistics thus provide an independent, local confirmation that memory effects act to suppress inertial clustering. Following Monchaux *et al.* (2010) and Monchaux (2012), cluster and void thresholds are identified as the intersections between the measured PDFs and the Poisson reference. Figure 14c shows that these thresholds separate further with increasing inertia, quantitatively reflecting growing intermittency.

As shown by Monchaux *et al.* (2010), clustering intensity is robustly characterized by the variance of $\mathcal{V}$. Figure 14d shows that for $St = 1$ the variance increases by a factor of four to five relative to the Poisson reference, consistent with the large RDF values and reduced correlation dimensions. At fixed St, increasing the density ratio systematically amplifies the variance.

The near-lognormal character of the Voronoï volume distribution is examined by considering the centered and normalized PDFs of $\log \mathcal{V}$ (figure 15a), which collapse satisfactorily onto a Gaussian distribution. This behavior is further supported by verification of the lognormal moment relations

$$\log\langle \mathcal{V} \rangle = \langle \log \mathcal{V} \rangle + \frac{1}{2}\sigma^2_{\log \mathcal{V}}, \tag{34}$$

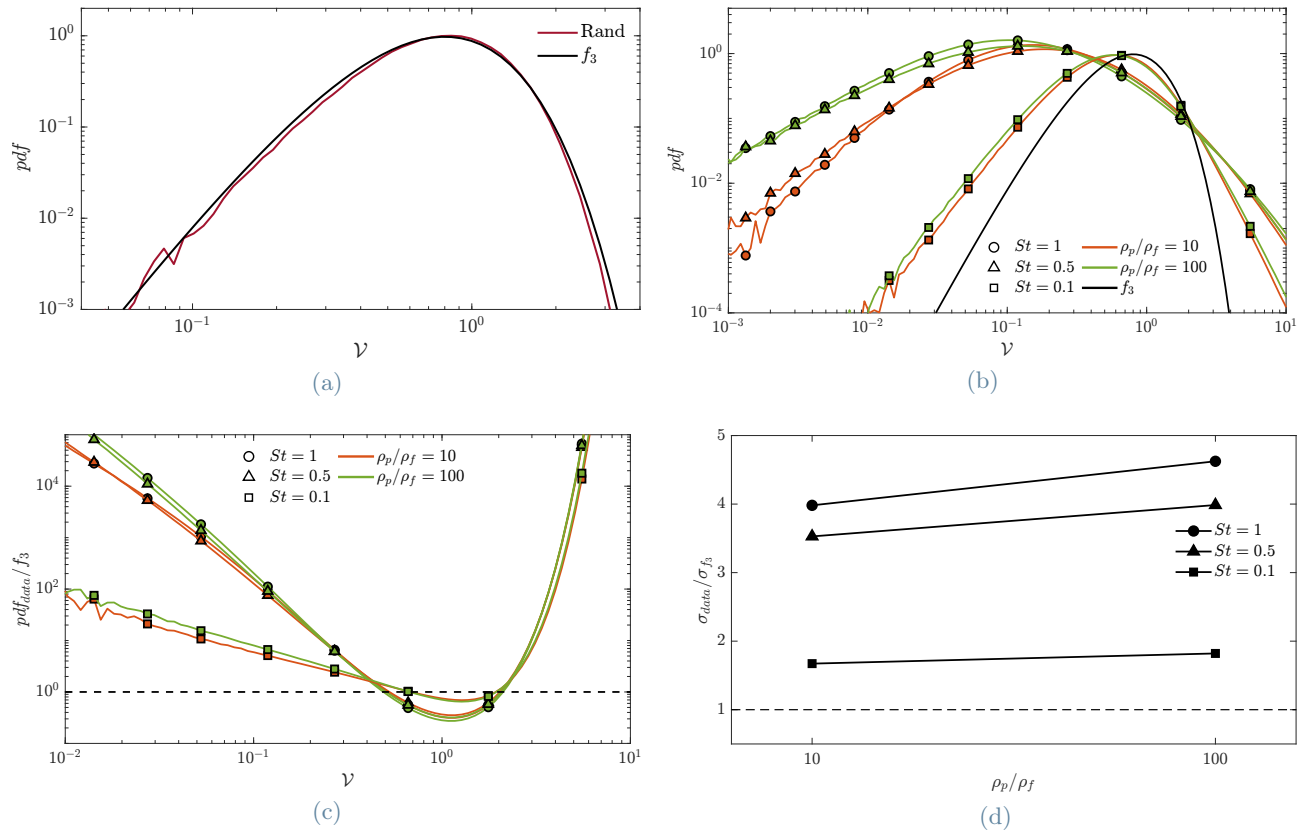


Figure 14: (a) Comparison between the analytical Poisson distribution f_3 and the numerical random reference. (b) PDFs of normalized Voronoï volumes $\mathcal{V}$ for all cases. (c) Ratio of the Voronoï volume PDFs to f_3, used to identify the clustering (left intersection with unity) and void (right intersection with unity) thresholds. (d) Ratio of the standard deviation of Voronoï volumes to the Poisson reference.

and

$$\sigma^2_{\mathcal{V}} = \langle \mathcal{V} \rangle^2 \left(e^{\sigma^2_{\log \mathcal{V}}} - 1 \right), \tag{35}$$

with $\langle \mathcal{V} \rangle = 1$ by construction. Both relations are satisfied with high accuracy, as shown in figure 15b.
In summary, the Voronoï analysis independently confirms the RDF and ADF results: strong inertial clustering is associated with enhanced intermittency and anisotropy, whereas weakly inertial particles remain close to a random Poisson distribution.

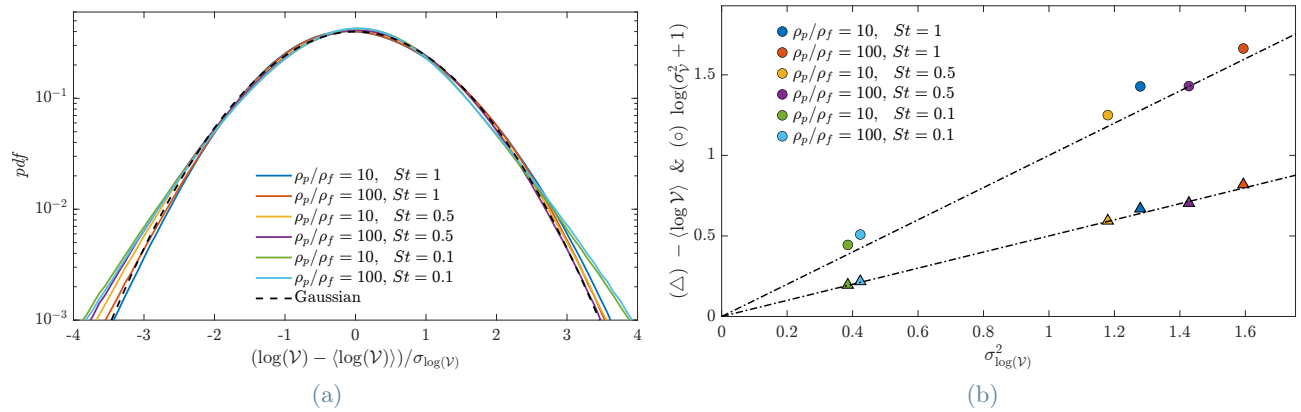


Figure 15: (a) Centered and normalized PDFs of $\log \mathcal{V}$ compared with a Gaussian distribution (black dashed line). (b) Verification of the lognormal relations (34) (triangles) and (35) (circles); lognormal behavior corresponds to the dash-dotted lines.

6. Structure, scaling, and orientation of inertial-particle clusters

The Voronoï tessellation not only identifies clustered particles but also provides a natural framework to extract the geometry, topology, and scaling properties of the clusters themselves. In this section, we analyze the size, shape, internal structure, and orientation of inertial-particle clusters in HST, with particular emphasis on their multiscale and fractal features and on the role of particle inertia and memory effects.
Clusters are defined as connected groups of particles whose Voronoï cell volumes fall below the lower intersection V^* between the measured distribution and the Poisson reference (figure 14). To avoid spurious connections due to tenuous links between isolated cells, we adopt the connectivity criterion proposed by Zamansky *et al.* (2016) and later used by Petersen *et al.* (2019): a particle is considered part of a cluster only if all its neighboring Voronoï cells also satisfy $V_V < V^*$. Clusters composed of fewer than four particles are discarded, as their geometric properties are not meaningful for the present analysis.

6.1. Cluster fractality

Self-similar and fractal characteristics of inertial-particle clusters have been widely documented in HIT (Goto & Vassilicos, 2006; Yoshimoto & Goto, 2007; Baker *et al.*, 2017; Petersen *et al.*, 2019).

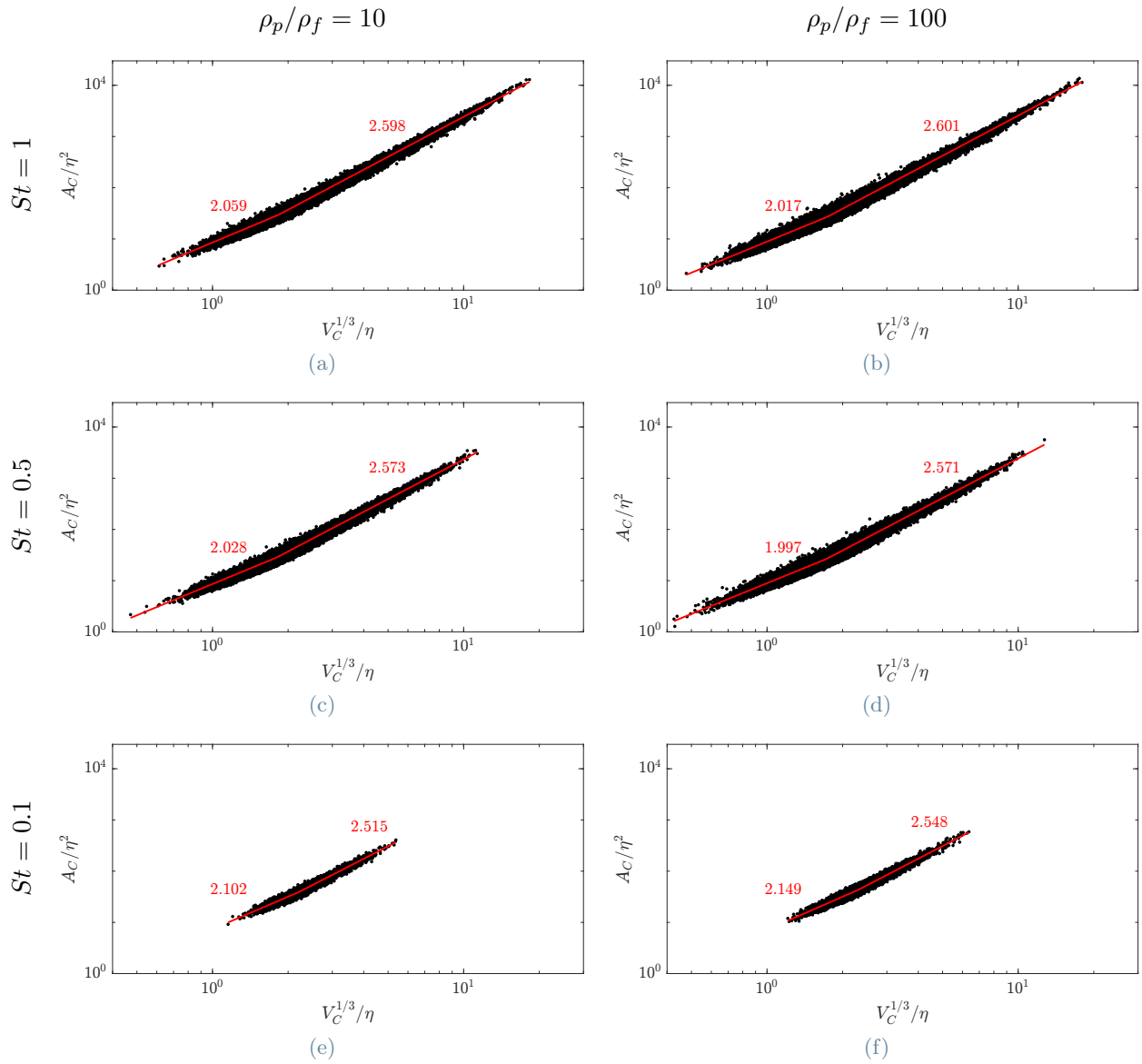


Figure 16: Scatter plot of cluster surface area A_C versus the cubic root of cluster volume $V_C^{1/3}$, both normalized by Kolmogorov scaling. Rows correspond to different Stokes numbers ($St = 1$, 0.5, 0.1 from top to bottom), and columns to different density ratios ($\rho_p/\rho_f = 10$ and 100).

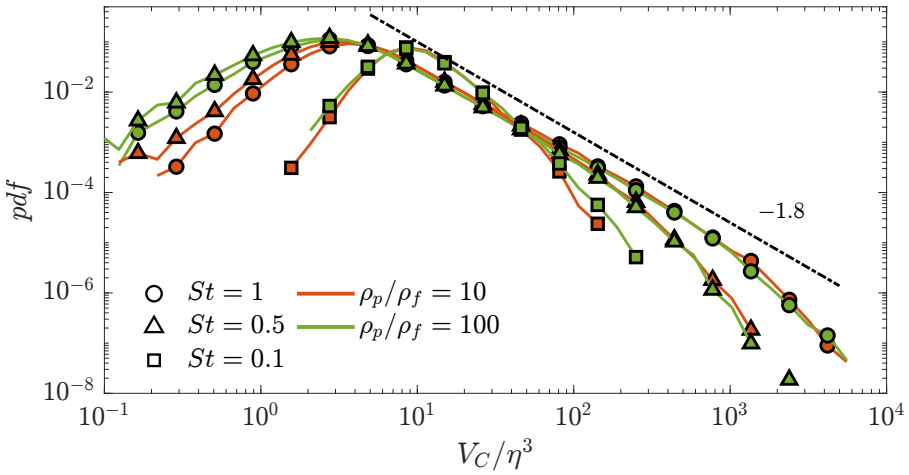


Figure 17: Probability density functions of particle cluster volumes, normalized by the Kolmogorov length scale. The dashed-dotted line indicates the power-law scaling observed in the self-similar region.

To assess whether these properties persist in the presence of mean shear, we examine the relation between cluster surface area A_C and the cubic root of cluster volume $V_C^{1/3}$, both normalized by Kolmogorov scales. This diagnostic provides a compact measure of surface roughness and geometric complexity across scales.

Figure 16 shows that small clusters follow the Euclidean scaling $A_C \sim V_C^{2/3}$, consistent with nearly compact three-dimensional objects. As cluster size increases, the scaling steepens, indicating increasingly convoluted surfaces and the emergence of fractal geometry. The resulting exponent, ranging between 2.5 and 2.6 across all cases, is systematically lower than the values exceeding 2.8 reported for HIT (Baker *et al.*, 2017). This suggests that mean shear smooths cluster surfaces and constrains geometric complexity, while leaving the overall clustering mechanism intact.

The transition from Euclidean to fractal scaling occurs at volumes of order $(1.8\eta)^3$ for $St = 1$ and $St = 0.5$, and $(2.3\eta)^3$ for $St = 0.1$. Following Baker *et al.* (2017), clusters above this threshold are referred to as *coherent clusters*, as their sizes reflect aggregation driven by coherent turbulent structures (Goto & Vassilicos, 2006; Yoshimoto & Goto, 2007). Smaller clusters, although connected, lack clear self-similarity.

Highly inertial cases ($St = 1$ and $St = 0.5$) display a pronounced broadening of cluster sizes: particles not only form denser aggregates but also participate in much larger connected structures spanning multiple scales.

Figure 17 reinforces this picture by showing the PDFs of cluster volumes. For volumes exceeding the coherent-cluster threshold, all cases exhibit a clear power-law decay, indicative of scale-invariant statistics. For $St = 1$ and $St = 0.5$, the slope approaches -1.8, consistent with void-volume scalings in HIT (Yoshimoto & Goto, 2007) and with the -2 exponent reported for particle clusters in both experiments and simulations (Monchaux *et al.*, 2010; Petersen *et al.*, 2019). The persistence of this scaling across different flows indicates that the formation of large inertial clusters is governed by robust turbulent mechanisms that are only weakly affected by mean shear. The power-law tail implies a non-negligible probability of clusters reaching inertial-range sizes (Petersen *et al.*, 2019). Our results therefore confirm that, while mean shear alters cluster geometry fractality, it does not suppress the formation of large or system-spanning aggregates.

6.2. Cluster topology

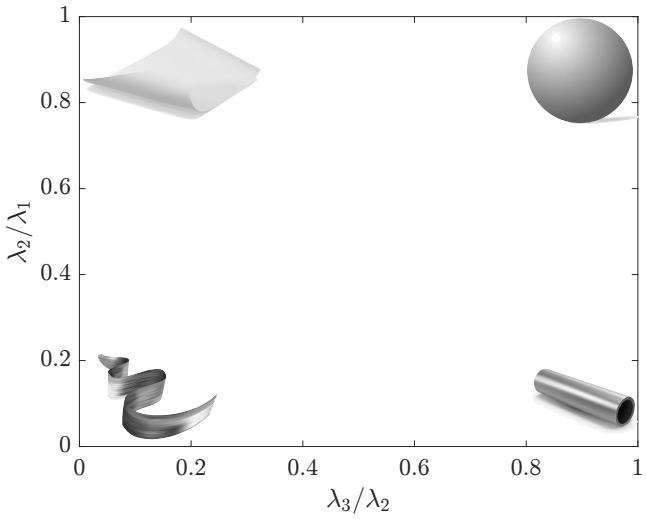
Beyond size and scaling, the Voronoï-based framework enables a detailed geometric characterization of cluster topology. Cluster shape is quantified using the gyration tensor. For a cluster composed of N particles at positions $\boldsymbol{x}^{(n)}$:

$$G_{ij} = \frac{1}{N}\sum_{n=1}^{N}\left(x_i^{(n)} - x_i^{cm}\right)\left(x_j^{(n)} - x_j^{cm}\right), \qquad \boldsymbol{x}^{cm} = \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}^{(n)}.$$

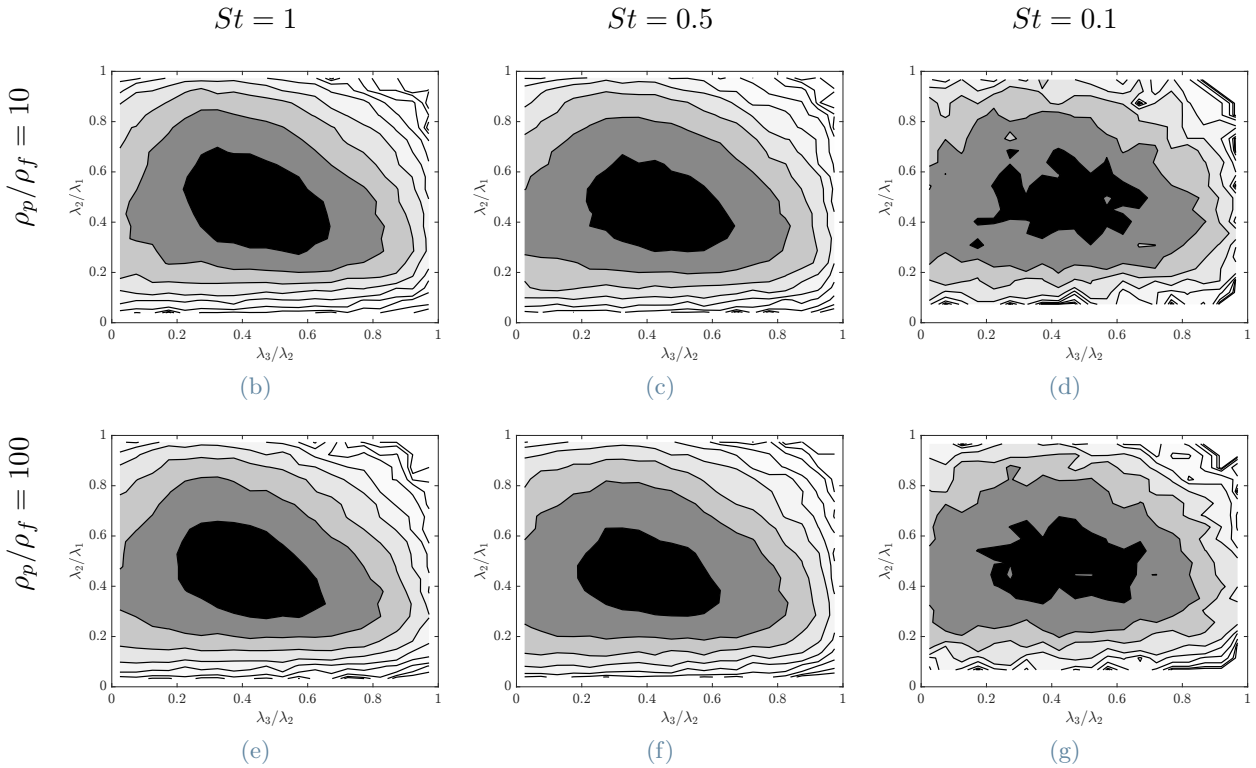
Being symmetric and positive semi-definite, G_{ij} can be diagonalized to yield principal radii $\lambda_1 \geq \lambda_2 \geq \lambda_3$, which quantify the spatial extent of the cluster along three orthogonal principal axes. The squared radius of gyration $R_g^2 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$ measures overall cluster size.

From these eigenvalues we define dimensionless shape descriptors,

$$b = \lambda_1^2 - \frac{1}{2}\left(\lambda_2^2 + \lambda_3^2\right), \qquad c = \lambda_2^2 - \lambda_3^2, \qquad k^2 = \frac{3}{2}\frac{\lambda_1^4 + \lambda_2^4 + \lambda_3^4}{\left(\lambda_1^2 + \lambda_2^2 + \lambda_3^2\right)^2} - \frac{1}{2},$$



(a)



(b) (c) (d)

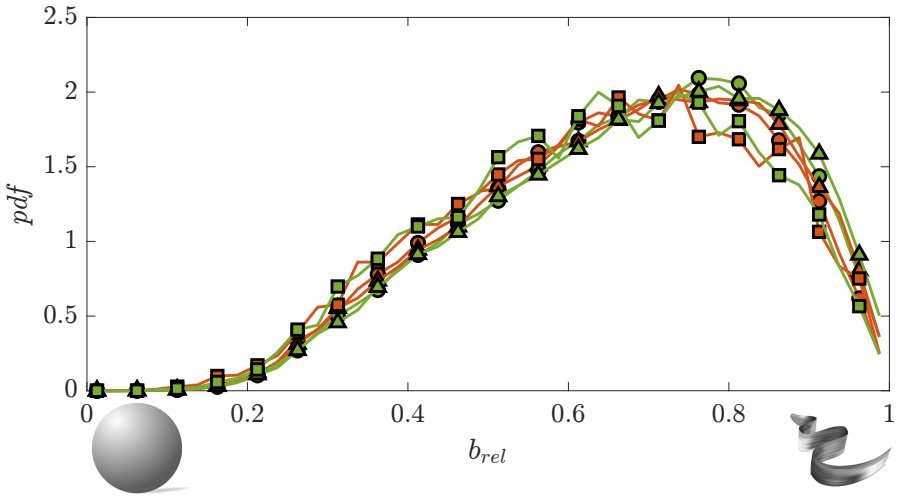
(e) (f) (g)

Figure 18: (a) Limiting cluster shapes inferred from the ratios of the eigenvalues of the gyration tensor (ribbon, sheet, tube, and sphere). (b-g) Joint distributions of the aspect ratios (λ_3/λ_2, λ_2/λ_1) for all cases with 10 contour levels from 5×10^{-3} to 5. Columns correspond to different Stokes numbers ($St = 1$, 0.5, 0.1 from left to right), and rows to different density ratios ($\rho_p/\rho_f = 10$ and 100).

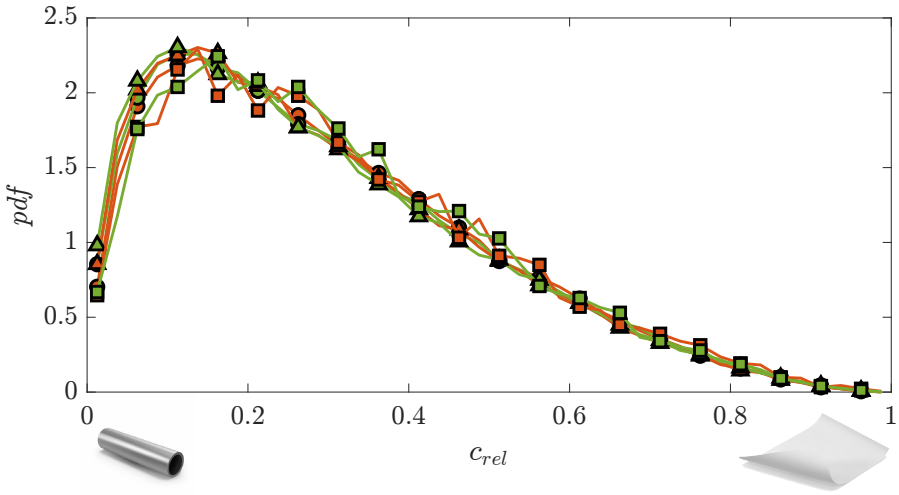
which measure asphericity, acylindricity, and relative anisotropy, respectively. The asphericity b quantifies deviations from spherical symmetry ($b = 0$ for a sphere), while the acylindricity c measures departures from circular symmetry in the plane orthogonal to the major axis. The relative anisotropy k^2 characterizes overall shape symmetry, vanishing for highly symmetric objects and approaching unity for collinear configurations.
Joint probability distribution functions (JPDFs) of the aspect ratios λ_3/λ_2 and λ_2/λ_1 are shown in figure 18. The limiting shapes correspond to

$$(\lambda_2/\lambda_1, \lambda_3/\lambda_2) = (0,0) \text{ ribbon}, \quad (1,0) \text{ sheet}, \quad (0,1) \text{ tube}, \quad (1,1) \text{ sphere}.$$

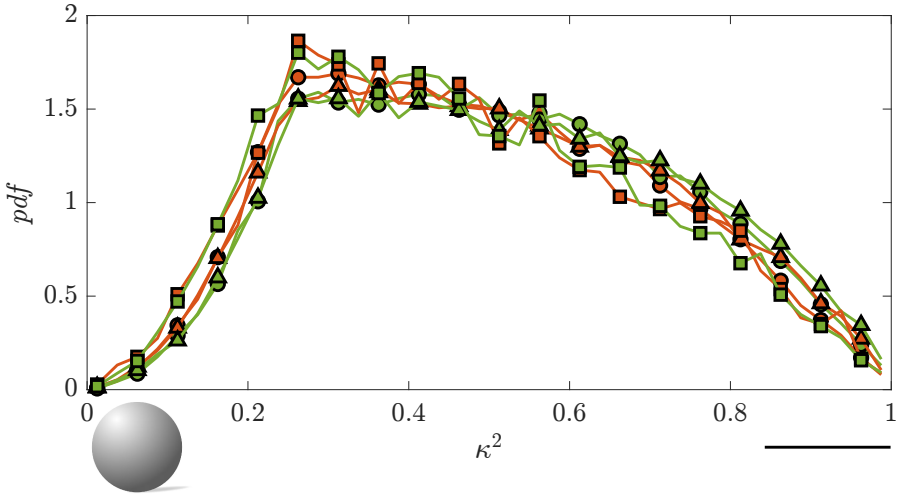
The JPDFs concentrate near $(\lambda_3/\lambda_2, \lambda_2/\lambda_1) \approx (0.5, 0.5)$, indicating moderately elongated, fully three-dimensional clusters. Remarkably, the distributions are nearly identical across Stokes numbers and density ratios, in agree-



(a)



(b)



(c)

Figure 19: Probability density functions of (a) normalized asphericity b/R_g^2, (b) normalized acylindricity $2c/R_g^2$, and (c) shape anisotropy k^2 for all cases. Illustrations below each x-axis indicate the typical cluster shape corresponding to the variable's value.

ment with HIT studies (Baker *et al.*, 2017; Petersen *et al.*, 2019). This universality suggests that intrinsic cluster topology is largely insensitive to to mean shear, flow anisotropy, or memory effects.

Consistent with this observation, the PDFs of normalized asphericity, acylindricity, and relative anisotropy (figure 19) collapse across all cases. Clusters are strongly non-spherical but not purely filamentary: cross-sections are moderately elliptical, and $k^2 \approx 0.3$–0.4 indicates elongated structures with substantial transverse deformation. Overall, inertial clusters form robust, cigar-like objects with a universal three-dimensional topology.

6.3. Cluster orientation

While cluster topology is universal, cluster orientation directly reflects the anisotropy induced by mean shear. Figure 20b shows the PDF of the polar angle θ_C between the cluster major axis and the vertical direction. All cases exhibit a clear preference for orientations between $\theta_C \approx \pi/4$ and $\pi/2$, indicating alignment within the shear plane.

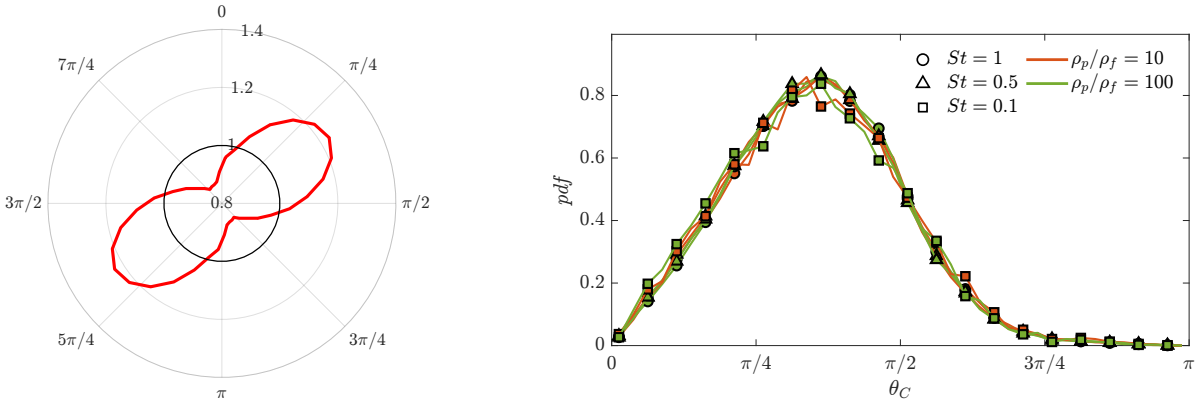


Figure 20: (a) Polar plot of the normalized ADF at $r \approx 5\eta$ for ($\rho_p/\rho_f = 10$, $St = 1$), highlighting preferential directions. (b) Probability density functions of cluster orientation angle θ_C between the cluster major axis and vertical.

The observed orientation closely mirrors the angular organization of particle pairs revealed by the ADF (figure 20a), demonstrating that anisotropy is coherent across scales. The mean cluster orientation, $\theta_C = 64° \pm 26°$, is distinctly more horizontal than the 45° extensional axis of the mean shear. This shift is consistent with the preferential angles reported by Lee & Lee (2020) for $St = 1$ particles in zero-gravity conditions.
Further support for this interpretation comes from Dong *et al.* (2017), who showed that coherent vortices in HST exhibit inclination angles of $52° \pm 8°$ at comparable Reynolds numbers. The close correspondence between vortex and cluster orientations indicates that inertial-particle clusters inherit their alignment from the underlying coherent flow structures rather than from the mean shear alone.

6.4. Cluster concentration

Finally, we examine how particle concentration varies with cluster size. The mean concentration of a cluster, $C_C = N_{PC}/V_C$, is evaluated as a function of cluster volume. Figure 21 shows that V_C scales almost linearly with the number of particles N_{PC} for all cases, with exponents in the range 1.04–1.10. This scaling implies that clusters maintain an approximately uniform internal concentration across sizes.
Deviations from linearity are confined to the smallest clusters, where Voronoï-based volume estimates are less reliable. For $St = 1$, the linear scaling spans nearly three decades in N_{PC}, reflecting the broad scale-invariance of inertial clustering. In contrast, for weakly inertial particles ($St = 0.1$), the linear regime is restricted to small clusters.
Overall, these results show that particle inertia primarily controls cluster extent and intermittency, while the internal concentration remains approximately uniform. Combined with fractal geometry and universal topology, this scale-invariant concentration demonstrates that inertial clustering in HST arises from intrinsic particle-flow interactions rather than from simple accumulation or local crowding effects.

7. Particles preferential sampling

This section investigates the physical mechanisms underlying inertial-particle clustering in HST. While preferential concentration has traditionally been interpreted through the centrifugal mechanism, whereby particles

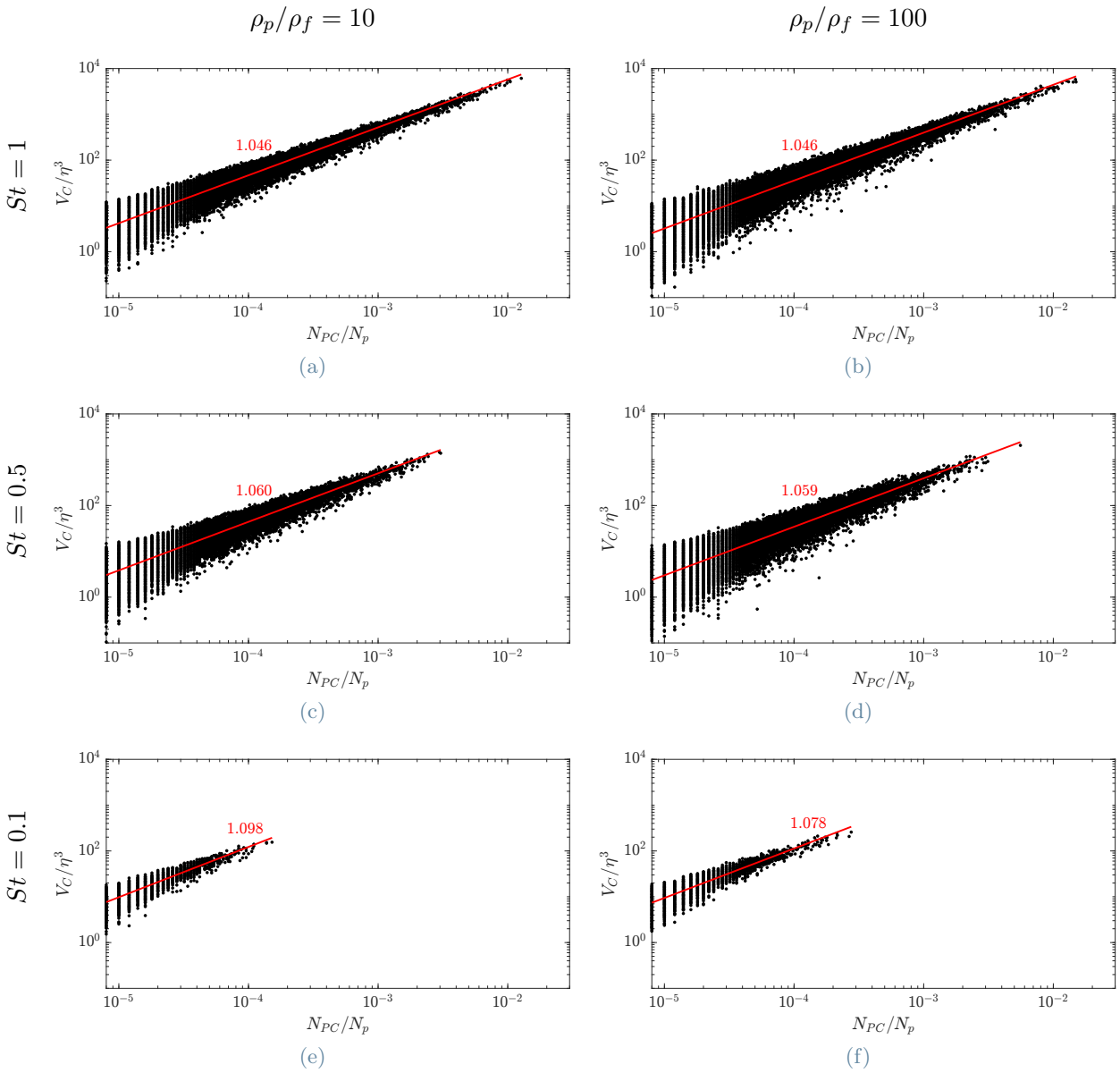
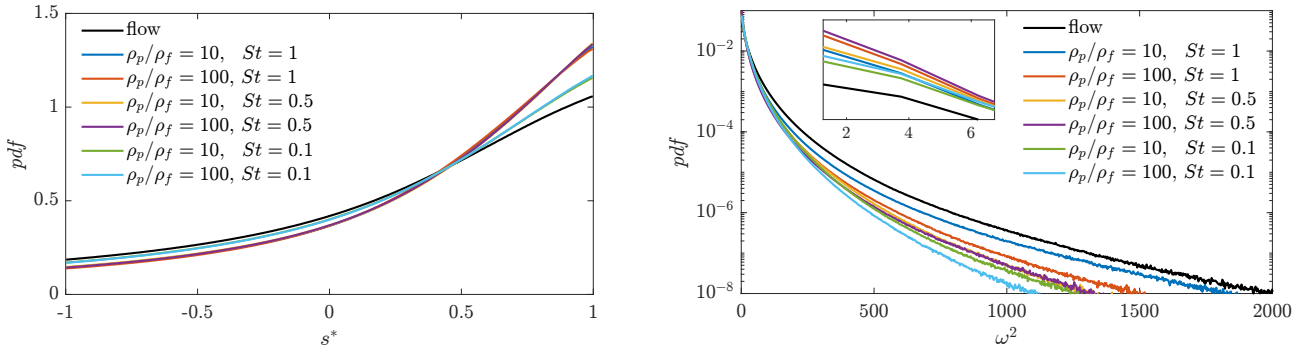


Figure 21: Cluster volume V_C (normalized by η^3) plotted against cluster particle count N_{PC} (normalized by the total number of particles). Rows correspond to different Stokes numbers ($St = 1$, 0.5, 0.1 from top to bottom), and columns to different density ratios ($\rho_p/\rho_f = 10$ and 100). Red lines denote best-fit power laws, showing an almost linear relation between V_C and N_{PC} for all cases.

are expelled from vortical regions, recent studies have highlighted the complementary role of the sweep-stick mechanism, particularly for weakly inertial particles. A systematic analysis of preferential sampling is therefore required to fully interpret the clustering statistics presented in the previous sections and to assess how memory effects modulate particle-flow interactions.

7.1. Structure of the flow sampled by particles

Building on the carrier-flow topology analysis in section 3.2, we first examine the velocity-gradient tensor sampled along particle trajectories. Attention is focused on the strain-state parameter s^* and the enstrophy ω^2, which together characterize the local morphology and intensity of the flow experienced by the particles.
Figure 22 shows that inertial particles explore a restricted subset of velocity-gradient states. The distribution of s^* (figure 22a) exhibit a pronounced bias toward $s^* \approx 1$, corresponding to axisymmetric extensional strain. This bias strengthens with increasing Stokes number, consistent with the enhanced clustering revealed by the RDF and Voronoï analyses for more inertial particles.
The enstrophy PDFs (figure 22b) display a clear depletion of high-ω^2 events, together with an enhanced prob-



(a) (b)

Figure 22: (a) Probability density function of the strain-state parameter s^* sampled along particle trajectories. (b) Probability density function of the enstrophy ω^2 sampled by the particles; the inset provides a magnified view of the low-ω^2 region. In both panels, deviations from Eulerian statistics indicate biased exploration of the flow.

ability of sampling low-ω^2 regions. This behavior indicates that inertial particles preferentially avoid intense vortical structures and accumulate in strain-dominated regions. The inset highlights that the cases exhibiting the strongest RDF peaks, and thus the most pronounced small-scale clustering, also show the largest excess probability at low-ω^2, confirming the close link between vortex avoidance and particle aggregation. At the same time, the tails of the distributions reflect the intermittent nature of turbulence: although particles predominantly reside in low-ω^2 regions, they still experience sporadic bursts of intense vorticity. The statistics of these rare high-ω^2 events are strongly influenced by the Basset history force. By introducing a memory of past velocity-gradient fluctuations, the Basset history force smooths the rapid variations of the local flow experienced by the particle. This temporal filtering reduces the sharpness of the particle response to intense vortical and strain-dominated events, leading to a broadening of the distribution tails.

Further insight is obtained by examining the second and third invariants of the velocity-gradient tensor, Q and R, together with their decomposed contributions Q_S and Q_W associated with strain- and rotation-dominated regions. Figure 23 presents both marginal and joint distributions sampled along particle trajectories.

The marginal PDFs of Q and R (figures 23a and 23b) confirm that particles explore a limited subset of flow topologies. In all cases, the distributions of Q are negatively skewed, indicating preferential sampling of strain-dominated regions and avoidance of vortical ones. At fixed St, decreasing the density ratio ρ_p/ρ_f (thereby increasing the relative importance of the Basset history term) systematically broadens both distributions. This behavior is consistent with the trends observed previously: the Basset term weakens the segregation between strain- and rotation-dominated regions, partially restoring exposure to extreme flow topologies.

The joint Q–R distributions (figures 23c and 23d) show that weakly inertial particles ($St = 0.1$) strongly avoid the focal, vortex-dominated region ($Q > 0$, $\Delta < 0$), while preferentially sampling biaxial strain-dominated states ($Q < 0$, $R > 0$). As particle inertia and memory effects increase, the probability lobes progressively extend into the vortex-stretching quadrants, reflecting a partial re-exposure to extreme flow topologies.

This behavior is reinforced by the joint distributions of Q_S and Q_W (figures 23e and 23f). Particles with weak memory effects avoid strongly rotation-dominated states ($Q_W \gg -Q_S$), whereas the inclusion of the Basset term allows broader sampling of such regions. Nevertheless, purely rotation-dominated states remain under-sampled, confirming that centrifugal effects remain active.

7.2. Centrifugal and sweep-stick mechanism

To further disentangle the mechanisms underlying clustering, we analyze the spatial organization of particles relative to coherent vortices and regions of weak fluid acceleration, following Uhlmann & Chouippe (2017).

7.2.1 Centrifugal mechanism

The centrifugal mechanism predicts particle expulsion from intense vortical regions and accumulation in strain-dominated zones. Coherent vortices are identified using the Q-criterion (Hunt *et al.*, 1988), with vortical regions defined by

$$Q > Q_{\text{thresh}} = 1.5\,\sigma_Q,$$

where σ_Q is the standard deviation of Q.

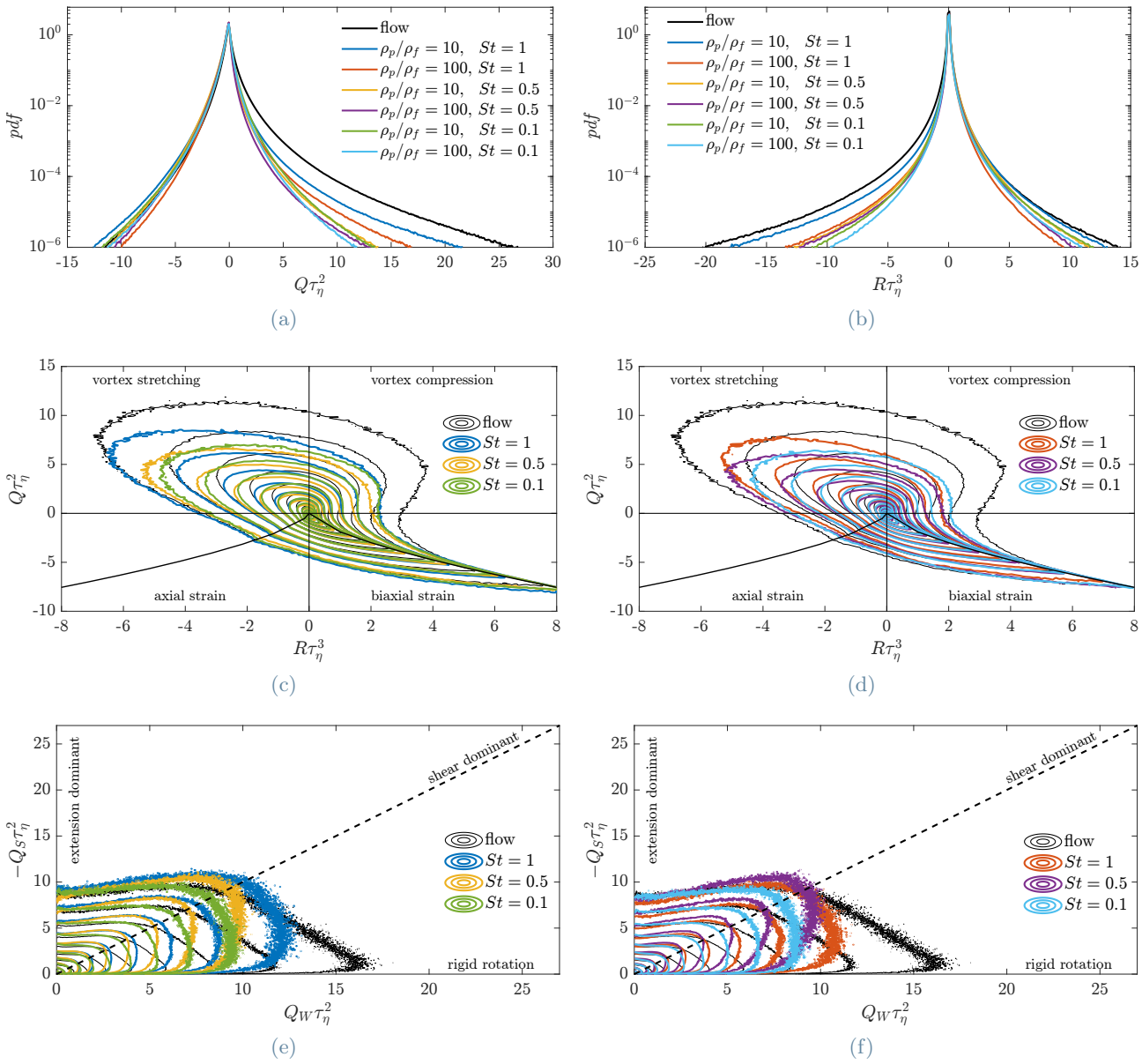
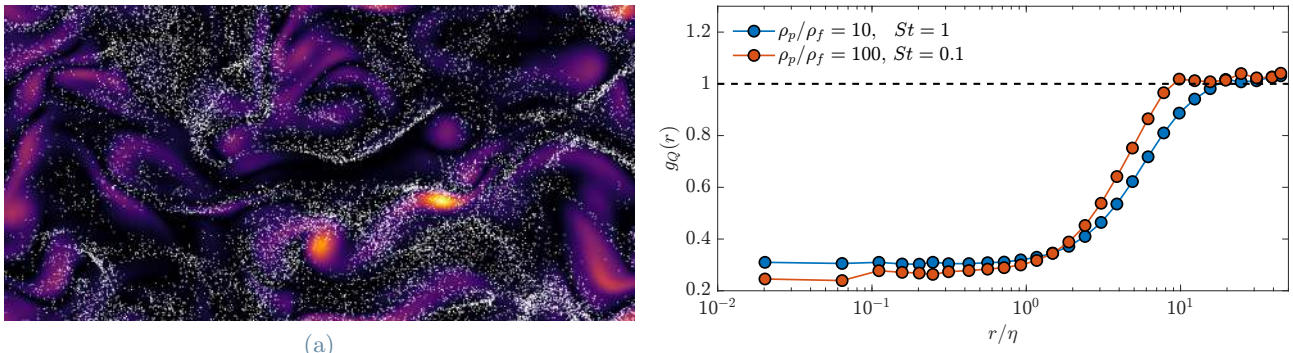


Figure 23: Marginal probability density functions of (a) Q and (b) R. Joint probability density functions in the Q–R plane for all Stokes numbers at density ratios (c) $\rho_p/\rho_f = 10$ and (d) 100. Joint probability density functions in the Q_S–Q_W plane for all Stokes numbers at density ratios (e) $\rho_p/\rho_f = 10$ and (f) 100.

Instantaneous particle configurations (figures 9 and 24a) reveal a clear depletion of particles within vortical cores. This effect is quantified by computing the radial distribution function between particle positions and Eulerian grid points located in high-Q regions ($Q > Q_{\text{thresh}}$), $g_Q(r)$.
Figure 24b shows that at separations of order the Kolmogorov scale, $g_Q(r) < 1$, confirming particle avoidance of intense vortical regions. Comparison between representative cases, ($\rho_p/\rho_f = 10$, $St = 1$) and ($\rho_p/\rho_f = 100$, $St = 0.1$), highlights the role of memory effects. When the Basset history force is negligible, depletion is stronger, reflecting efficient centrifugal ejection. Conversely, when memory effects are significant, depletion is slightly reduced, indicating that particles penetrate vortical regions more frequently. This behavior aligns with the preferential-sampling results, which show that memory effects weaken segregation by smoothing rapid fluctuations of the velocity gradient.

7.2.2 Sweep-stick mechanism

The sweep-stick mechanism describes the tendency of inertial particles to accumulate in regions of weak fluid acceleration, where they can remain trapped for extended periods. Within the one-way coupled point-particle



(a) (b)

Figure 24: (a) Magnification of the instantaneous particle configuration in the x–y plane for the case ($\rho_p/\rho_f = 10$, $St = 1$), superimposed to the modulus of the vorticity component normal to the plotted plane. (b) Radial distribution function between particles and Eulerian grid points in intense vortical regions ($Q > Q_{\text{thresh}}$). Values below unity indicate particle depletion near vortices.

framework, Chen *et al.* (2006); Goto & Vassilicos (2008) showed that clustering occurs near special points satisfying

$$\boldsymbol{e}_1 \cdot \boldsymbol{a}_f = 0, \qquad \lambda_1 > 0, \tag{36}$$

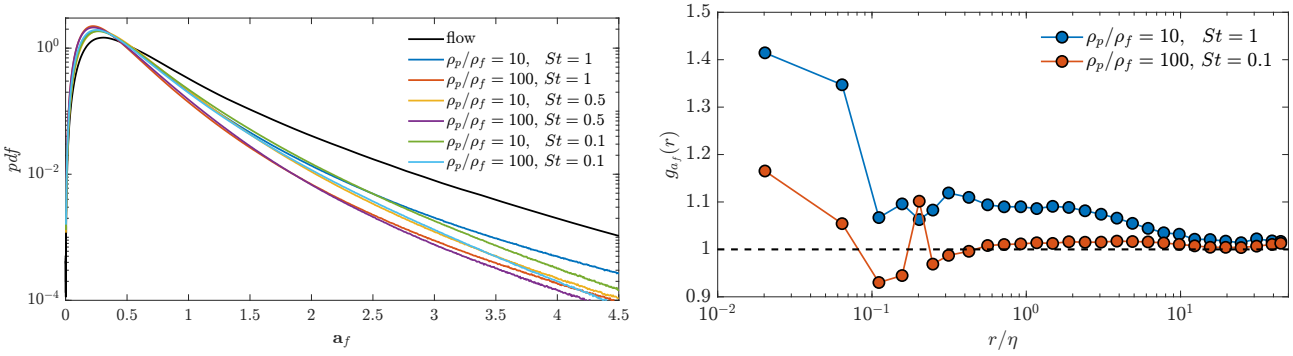
where λ_1 is the largest eigenvalue of the symmetric part of the acceleration-gradient tensor and $\boldsymbol{e}_1$ is the corresponding eigenvector. Physically, for small inertial particles (in a simplified MRG model) the slip velocity scales as

$$\boldsymbol{u}_p - \boldsymbol{u}_f \simeq -\tau_p \boldsymbol{a}_f,$$

so that particles drift toward regions where the fluid acceleration vanishes. Once near these "sticky" points, particles can remain trapped as long as the local flow topology persists.
Motivated by the findings of Coleman & Vassilicos (2009), we examine both the magnitude of the fluid acceleration and the more restrictive sweep-stick condition. The fluid acceleration,

$$\boldsymbol{a}_f = \frac{D\boldsymbol{u}}{Dt} = \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u},$$

is evaluated along particle trajectories and throughout the Eulerian field.



(a) (b)

Figure 25: (a) Probability density functions of the magnitude of the fluid acceleration $|\boldsymbol{a}_f|$ sampled along particle trajectories for different density ratios and Stokes numbers. The black curve shows the Eulerian reference statistics. (b) Radial distribution function $g_{a_f}(r)$ between particle positions and Eulerian grid points satisfying the sweep-stick condition defined in equation (37).

Figure 25a shows that particles preferentially sample regions of weak acceleration, with a clear depletion of intense acceleration events relative to Eulerian statistics. This behavior is consistent with the sweep-stick picture. The breadth of the sampled acceleration distributions depends strongly on the particle-to-fluid density ratio. Lighter particles ($\rho_p/\rho_f = 10$) sample a broader range of acceleration magnitudes, reflecting stronger

sensitivity to acceleration fluctuations. Heavier particles ($\rho_p/\rho_f = 100$) remain more tightly concentrated near low-acceleration regions, resulting in narrower distributions and stronger depletion of high-acceleration tails. Within the explored parameter space, the dependence on Stokes number is comparatively weaker.
This trend can be understood from the MRG equation. The pressure gradient and added mass terms scale with the fluid acceleration, and their relative importance increases as the density ratio decreases. For the present parameters, the added mass contribution is negligible, while the pressure gradient force becomes significant for lighter particles. As a result, lighter particles respond more strongly to acceleration fluctuations, whereas heavier particles preferentially remain near low-acceleration regions.
To directly test the original sweep-stick condition, we compute the RDF between particle positions and Eulerian grid points satisfying

$$\boldsymbol{e}_1 \cdot \boldsymbol{a}_f < 4 \times 10^{-4}\, \sigma_{a_f}, \tag{37}$$

where σ_{a_f} is the standard deviation of $\boldsymbol{e}_1 \cdot \boldsymbol{a}_f$ (Uhlmann & Chouippe, 2017). The resulting RDF, $g_{a_f}(r)$, is shown in figure 25b. At large separations, $g_{a_f}(r) \to 1$, indicating no long-range correlations, while at small separations $g_{a_f}(r) > 1$, confirming preferential residence near sticky points.
The modest amplitude of $g_{a_f}(r)$ reflects the small fraction of grid points satisfying the stringent sweep-stick criterion. Nevertheless, the signal is systematic and most pronounced when particle response times are comparable to the Kolmogorov time scale, in agreement with theoretical expectations (Uhlmann & Chouippe, 2017).
Taken together, figures 25a and 25b show that particle clustering in HST results from the coexistence of multiple mechanisms. Preferential sampling of low-acceleration regions produces a robust global signature, while the sweep-stick condition identifies localized trapping sites whose effect, though weaker, is detectable. This coexistence highlights the complexity of inertial-particle dynamics governed by the full MRG equation and extends beyond the simplified sweep-stick framework originally derived for idealized flows.

8. Conclusions

We have investigated inertial-particle clustering in homogeneous shear turbulence using direct numerical simulations coupled with a Lagrangian particle tracker. The study is based on one-way coupled, point-particle direct numerical simulations of small particles, whose motion is described by the complete Maxey–Riley–Gatignol equation (Maxey & Riley, 1983; Gatignol, 1983), including the Basset history term and the Faxén corrections. The objective of the study are fourfold: *(i)* to quantify the relative contributions of each term in the Maxey–Riley–Gatignol equation in homogeneous shear turbulence; *(ii)* to assess the impact of the Basset history term on the intensity and anisotropy of particle clustering; *(iii)* to characterize the geometrical topology of clusters and identify potential universal features; and *(iv)* to determine the dominant mechanisms driving the clustering process. To achieve this, we explored a broad parameter space spanning different particle-to-fluid density ratios and particle inertia, expressed via the Stokes number. The micro-scale Reynolds number is $Re_\lambda \approx 47$, providing a sufficiently wide range of active scales to examine the influence of flow scales on clustering dynamics.

The analysis of the individual contributions in the Maxey–Riley–Gatignol equation shows that the relative importance of the Basset history term with respect to the Stokes drag scales with particle size: the larger the particle, the stronger the influence of memory effects. When the Stokes number is held fixed, variations in the particle-to-fluid density ratio effectively mask the role of particle size itself, so that no independent density-ratio effect is observed. This analysis further demonstrates that simplified particle-dynamics models neglecting the Basset history term remain accurate for sufficiently small particles, with characteristic sizes of order 10^{-2} the Kolmogorov scale. We also examined the relative magnitude of all force terms and found that the scaling estimates proposed by Daitche (2015) accurately capture, across almost all cases considered, the mean contribution of each term to particle dynamics. These estimates therefore provide a reliable basis for reduced-order models aimed at reproducing the mean particle behavior.
The radial distribution function and the angular distribution function were employed to quantify both the intensity of particle-particle distribution and its anisotropic character. The results are consistent with previous observations in homogeneous isotropic and shear turbulence (Gualtieri *et al.*, 2009; Olivieri *et al.*, 2014): particles with Stokes numbers of order unity exhibit enhanced small-scale clustering, whereas weakly inertial particles behave similarly to tracers and remain nearly homogeneously distributed. The inclusion of the Basset history term systematically weakens clustering. For the largest particles, for which memory effects are strongest, it leads to an earlier saturation of the radial distribution function, effectively acting as a suppressor of extreme clustering events. Further insight is provided by the spherical-harmonic decomposition of the angular distribution function, which reveals that the particle distribution is dominated by a quadrupolar mode aligned along the bottom-left to top-right diagonal of the mean vorticity-shear (x–z) plane. The Basset history term again reduces the strength of this angular dependence, and the saturation of the radial distribution function is accompanied by

a corresponding saturation of anisotropy, indicating a close coupling between radial clustering and angular organization.

The Voronoï tessellation (Monchaux *et al.*, 2010; Monchaux, 2012), in which each particle is associated with the region of space closer to it than to any other particle, was employed to directly access cluster geometry, topology, and scaling properties. This analysis first confirms the trends observed in the radial distribution function: particles with higher inertia exhibit strongly skewed Voronoï volume distributions, with an enhanced probability of small cell volumes compared with a random Poisson process, reflecting intense clustering. Following Zamansky *et al.* (2016), clusters were identified as connected sets of particles whose Voronoï cell volumes fall below a threshold defined by the lower intersection between the measured distribution and the Poisson reference, with the additional requirement that all neighboring particles satisfy the same condition. Small clusters are found to be quasi-Euclidean, whereas larger clusters exhibit increasingly convoluted and fractal geometries, in close analogy with observations in homogeneous isotropic turbulence (Baker *et al.*, 2017). The characterization of the cluster gyration tensor further allowed us to investigate cluster topology and spatial orientation. The results reveal a remarkable degree of universality in the clustering process: cigar-like clusters are the most probable configuration for both highly inertial particles and weakly inertial, quasi-Lagrangian tracers, characterized by strong elongation along a principal axis and a moderately circular cross-section. Cluster orientation is preferentially more horizontal than the shear direction, indicating that cluster alignment is governed by the interplay between large-scale shear-induced anisotropy and coherent vortical structures in the carrier flow.

We finally investigated preferential sampling and the physical mechanisms underlying particle clustering. The small-scale topology of both the carrier flow and the particle-sampled flow was analyzed through the invariants of the velocity-gradient tensor $A_{ij} = \partial u_i/\partial x_j$ (Meneveau, 2011). Analysis of the strain-rate eigenvalues shows that particles preferentially concentrate in regions of axisymmetric extension. Joint probability density functions of the second and third invariants of A_{ij} further indicate enhanced sampling of biaxial strain states and a depletion in the focal region, consistent with an active centrifugal mechanism that expels inertial particles from vortical regions. The Basset history force acts as a temporal filter, damping extreme velocity fluctuations and allowing larger particles to explore a broader subset of the flow field. This behavior is further supported by the joint statistics of the second invariants of the strain-rate and rotation-rate tensors, as well as by radial distribution functions computed between particles and grid points conditioned on high values of the second invariant of A_{ij} (exceeding 1.5 standard deviations). The sweep-stick mechanism was examined through the probability density function of the flow acceleration magnitude and through radial distribution functions between particles and grid points satisfying the sweep-stick condition proposed by Chen *et al.* (2006); Goto & Vassilicos (2008). Inertial particles preferentially reside in regions where the flow acceleration is weak. This behavior exhibits only a limited sensitivity to the Basset history force and is instead primarily governed by the particle-to-fluid density ratio. The force term in the Maxey–Riley–Gatignol equation directly associated with the flow acceleration and inversely proportional to the density ratio is the pressure gradient contribution. Particles with lower density ratios, and therefore stronger pressure-gradient effects, sample a broader range of accelerations, in a manner analogous to the role of the history force in moderating centrifugal ejection.

The insights gained in this work open several directions for future research. Extending the analysis to higher Reynolds numbers, lower particle-to-fluid density ratios, polydisperse suspensions, and finite-size particles (e.g., by resolving the flow around each particle through direct numerical simulations of the Navier–Stokes equations coupled with the immersed boundary method) would help evaluate the robustness of the observed universality and clarify the role of additional physical effects, including particle back-reaction on the flow, thereby providing a direct test of the one-way coupled point-particle approximation. While the present results provide a detailed characterization of clustering intensity, anisotropy, and internal structure, their implications for collision rates, relative dispersion, and mixing remain largely unexplored. Further work is also needed to assess how memory effects and sweep-stick dynamics interact with flow anisotropy at higher Reynolds numbers and in more complex configurations, such as wall-bounded or stratified shear flows. In this context, the multi-scale framework developed here provides a flexible and systematic basis for addressing these open problems.

References

Aliseda, A., Cartellier, A., Hainaux, F. & Lasheras, J. C. 2002 Effect of preferential concentration on the settling velocity of heavy particles in homogeneous isotropic turbulence. *Journal of Fluid Mechanics* **468**, 77–105.

Ashurst, Wm. T., Kerstein, A. R., Kerr, R. M. & Gibson, C. H. 1987 Alignment of vorticity and scalar gradient with strain rate in simulated Navier–Stokes turbulence. *The Physics of Fluids* **30** (8), 2343–2353.

Auton, T. R., Hunt, J. C. R. & Prud'Homme, M. 1988 The force exerted on a body in inviscid unsteady non-uniform rotational flow. *Journal of Fluid Mechanics* **197**, 241–257.

Baker, L., Frankel, A., Mani, A. & Coletti, F. 2017 Coherent clusters of inertial particles in homogeneous turbulence. *Journal of Fluid Mechanics* **833**, 364–398.

Balachandar, S. & Eaton, J. K. 2010 Turbulent Dispersed Multiphase Flow. *Annual Review of Fluid Mechanics* **42** (1), 111–133.

Balachandar, S. & Maxey, M. R. 1989 Methods for evaluating fluid velocities in spectral simulations of turbulence. *Journal of Computational Physics* **83** (1), 96–125.

Balkovsky, E., Falkovich, G. & Fouxon, A. 2001 Intermittent Distribution of Inertial Particles in Turbulent Flows. *Physical Review Letters* **86** (13), 2790–2793.

Bec, J. 2003 Fractal clustering of inertial particles in random flows. *Physics of Fluids* **15** (11), L81–L84.

Bec, J., Biferale, L., Cencini, M., Lanotte, A., Musacchio, S. & Toschi, F. 2007 Heavy Particle Concentration in Turbulence at Dissipative and Inertial Scales. *Physical Review Letters* **98** (8), 084502.

Bec, J., Musacchio, S. & Ray, S. S. 2013 Sticky elastic collisions. *Physical Review E* **87** (6), 063013.

Bracco, A., Chavanis, P. H., Provenzale, A. & Spiegel, E. A. 1999 Particle aggregation in a turbulent Keplerian flow. *Physics of Fluids* **11** (8), 2280–2287.

Cantwell, B. J. 1993 On the behavior of velocity gradient tensor invariants in direct numerical simulations of turbulence. *Physics of Fluids A: Fluid Dynamics* **5** (8), 2008–2013.

Capecelatro, J., Pepiot, P. & Desjardins, O. 2015 Numerical investigation and modeling of reacting gas-solid flows in the presence of clusters. *Chemical Engineering Science* **122**, 403–415.

Casas, G., Ferrer, A. & Oñate, E. 2018 Approximating the Basset force by optimizing the method of van Hinsberg et al. *Journal of Computational Physics* **352**, 142–171.

Chambers, J. E. 2010 Planetesimal formation by turbulent concentration. *Icarus* **208** (2), 505–517.

Chen, L., Goto, S. & Vassilicos, J. C. 2006 Turbulent clustering of stagnation points and inertial particles. *Journal of Fluid Mechanics* **553**, 143–154.

Chiarini, A. & Rosti, M. E. 2024 Finite-size inertial spherical particles in turbulence. *Journal of Fluid Mechanics* **988**, A17.

Chong, M. S., Perry, A. E. & Cantwell, B. J. 1990 A general classification of three-dimensional flow fields. *Physics of Fluids A: Fluid Dynamics* **2** (5), 765–777.

Coleman, S. W. & Vassilicos, J. C. 2009 A unified sweep-stick mechanism to explain particle clustering in two- and three-dimensional homogeneous, isotropic turbulence. *Physics of Fluids* **21** (11), 113301.

Corrsin, S. 1958 Local isotropy in turbulent shear flow. Research Memorandum RM 58B11. National Advisory Committee for Aeronautics, Washington, D.C.

Daitche, A. 2015 On the role of the history force for inertial particles in turbulence. *Journal of Fluid Mechanics* **782**, 567–593.

Davidson, P. A. 2015 *Turbulence: An Introduction for Scientists and Engineers*, 2nd edn. Oxford: Oxford University Press.

De Lillo, F., Cencini, M., Durham, W. M., Barry, M., Stocker, R., Climent, E. & Boffetta, G. 2014 Turbulent Fluid Acceleration Generates Clusters of Gyrotactic Microorganisms. *Physical Review Letters* **112** (4), 044502.

Domingo, P, Vervisch, L & Reveillon, J 2005 DNS analysis of partially premixed combustion in spray and gaseous turbulent flame-bases stabilized in hot air. *Combustion and Flame* **140** (3), 172–195.

Dong, S., Lozano-Durán, A., Sekimoto, A. & Jiménez, J. 2017 Coherent structures in statistically stationary homogeneous shear turbulence. *Journal of Fluid Mechanics* **816**, 167–208.

Durham, W. M. & Stocker, R. 2012 Thin Phytoplankton Layers: Characteristics, Mechanisms, and Consequences. *Annual Review of Marine Science* **4** (1), 177–207.

Eaton, J. K. & Fessler, J. R. 1994 Preferential concentration of particles by turbulence. *International Journal of Multiphase Flow* **20**, 169–209.

Elghobashi, S. & Truesdell, G. C. 1993 On the two-way interaction between homogeneous turbulence and dispersed solid particles. I: Turbulence modification. *Physics of Fluids A: Fluid Dynamics* **5** (7), 1790–1801.

Falkovich, G., Fouxon, A. & Stepanov, M. G. 2002 Acceleration of rain initiation by cloud turbulence. *Nature* **419** (6903), 151–154.

Ferenc, J.-S. & Néda, Z. 2007 On the size distribution of Poisson Voronoi cells. *Physica A: Statistical Mechanics and its Applications* **385** (2), 518–526.

Gatignol, R. 1983 The Faxén formulae for a rigid particle in an unsteady non-uniform Stokes flow. *Journal de Mécanique Théorique et Appliquée* **2** (2), 143–160.

Goto, S. & Vassilicos, J. C. 2006 Self-similar clustering of inertial particles and zero-acceleration points in fully developed two-dimensional turbulence. *Physics of Fluids* **18** (11), 115103.

Goto, Susumu & Vassilicos, J. C. 2008 Sweep-Stick Mechanism of Heavy Particle Clustering in Fluid Turbulence. *Physical Review Letters* **100** (5), 054503.

Grabowski, W. W. & Wang, L.-P. 2013 Growth of Cloud Droplets in a Turbulent Environment. *Annual Review of Fluid Mechanics* **45** (1), 293–324.

Grassberger, P. & Procaccia, I. 1983 Characterization of Strange Attractors. *Physical Review Letters* **50** (5), 346–349.

Gualtieri, P., Casciola, C. M., Benzi, R., Amati, G. & Piva, R. 2002 Scaling laws and intermittency in homogeneous shear flow. *Physics of Fluids* **14** (2), 583–596.

Gualtieri, P., Picano, F. & Casciola, C. M. 2009 Anisotropic clustering of inertial particles in homogeneous shear flow. *Journal of Fluid Mechanics* **629**, 25–39.

Homann, H. & Bec, J. 2010 Finite-size effects in the dynamics of neutrally buoyant particles in turbulent flow. *Journal of Fluid Mechanics* **651**, 81–91.

Hunt, J. C. R., Wray, A. A. & Moin, P. 1988 Eddies, stream and convergence zones in turbulent flows. *Tech. Rep.* CTR-S88. Center for Turbulence Research.

Ireland, P. J., Bragg, A. D. & Collins, L. R. 2016 The effect of Reynolds number on inertial particle dynamics in isotropic turbulence. Part 1. Simulations without gravitational effects. *Journal of Fluid Mechanics* **796**, 617–658.

Jacobsen, J., Jayaraman, A. & Belmonte, Andrew 2000 Monotone Solutions of a Nonautonomous Differential Equation for a Sedimenting Sphere. *Electronic Journal of Differential Equations* **2001**.

Jeffrey, A. 2004 Fresnel Integrals, Sine and Cosine Integrals. In *Handbook of Mathematical Formulas and Integrals*, pp. 245–248. Elsevier.

Johansen, A., Oishi, J. S., Low, M.-M. M., Klahr, H., Henning, T. & Youdin, A. 2007 Rapid planetesimal formation in turbulent circumstellar disks. *Nature* **448** (7157), 1022–1025.

Károlyi, G., Péntek, Á., Scheuring, I., Tél, T. & Toroczkai, Z. 2000 Chaotic flow: The physics of species coexistence. *Proceedings of the National Academy of Sciences* **97** (25), 13661–13665.

Kim, J., Moin, P. & Moser, R. 1987 Turbulence statistics in fully developed channel flow at low Reynolds number. *Journal of Fluid Mechanics* **177**, 133–166.

Kleeorin, N. & Rogachevskii, I. 2025 Large-scale clustering of inertial particles in a rotating, stratified and inhomogeneous turbulence. *Physics of Fluids* **37** (6), 065152.

Lee, S. & Lee, C. 2020 Behavior of settling particles in homogeneous shear turbulence. *Physical Review Fluids* **5** (10), 104306.

Lees, A. W. & Edwards, S. F. 1972 The computer study of transport processes under extreme conditions. *Journal of Physics C: Solid State Physics* **5** (15), 1921–1928.

Luchini, P. 2020 Introducing CPL.

Luchini, P. & Quadrio, M. 2006 A low-cost parallel implementation of direct numerical simulation of wall turbulence. *Journal of Computational Physics* **211** (2), 551–571.

Lund, T. S. & Rogers, M. M. 1994 An improved measure of strain state probability in turbulent flows. *Physics of Fluids* **6** (5), 1838–1847.

Maxey, M. R. 1987 The gravitational settling of aerosol particles in homogeneous turbulence and random flow fields. *Journal of Fluid Mechanics* **174**, 441–465.

Maxey, M. R. & Riley, J. J. 1983 Equation of motion for a small rigid sphere in a nonuniform flow. *The Physics of Fluids* **26** (4), 883–889.

Meneveau, C. 2011 Lagrangian Dynamics and Models of the Velocity Gradient Tensor in Turbulent Flows. *Annual Review of Fluid Mechanics* **43** (1), 219–245.

Monchaux, R. 2012 Measuring concentration with Voronoï diagrams: The study of possible biases. *New Journal of Physics* **14** (9), 095013.

Monchaux, R., Bourgoin, M. & Cartellier, A. 2010 Preferential concentration of heavy particles: A Voronoï analysis. *Physics of Fluids* **22** (10), 103304.

Niazi Ardekani, M., Sardina, G., Brandt, L., Karp-Boss, L., Bearon, R. N. & Variano, E. A. 2017 Sedimentation of inertia-less prolate spheroids in homogenous isotropic turbulence with application to non-motile phytoplankton. *Journal of Fluid Mechanics* **831**, 655–674.

Nicolai, C., Jacob, B., Gualtieri, P. & Piva, R. 2014 Inertial Particles in Homogeneous Shear Turbulence: Experiments and Direct Numerical Simulation. *Flow, Turbulence and Combustion* **92** (1-2), 65–82.

Nomura, K. K. & Post, G. K. 1998 The structure and dynamics of vorticity and rate of strain in incompressible homogeneous turbulence. *Journal of Fluid Mechanics* **377**, 65–97.

Olivieri, S., Picano, F., Sardina, G., Iudicone, D. & Brandt, L. 2014 The effect of the Basset history force on particle clustering in homogeneous and isotropic turbulence. *Physics of Fluids* **26** (4), 041704.

Ooi, A., Martin, J., Soria, J. & Chong, M. S. 1999 A study of the evolution and characteristics of the invariants of the velocity-gradient tensor in isotropic turbulence. *Journal of Fluid Mechanics* **381**, 141–174.

Petersen, A. J., Baker, L. & Coletti, Filippo 2019 Experimental study of inertial particles clustering and settling in homogeneous turbulence. *Journal of Fluid Mechanics* **864**, 925–970.

Portela, L. M., Cota, P. & Oliemans, R. V. A. 2002 Numerical study of the near-wall behaviour of particles in turbulent pipe flows. *Powder Technology* **125** (2-3), 149–157.

Prasath, S. G., Vasan, V. & Govindarajan, R. 2019 Accurate solution method for the Maxey–Riley equation, and the effects of Basset history. *Journal of Fluid Mechanics* **868**, 428–460.

Ravichandran, S. & Govindarajan, R. 2017 Vortex-dipole collapse induced by droplet inertia and phase change. *Journal of Fluid Mechanics* **832**, 745–776.

Reeks, M. W. 1983 The transport of discrete particles in inhomogeneous turbulence. *Journal of Aerosol Science* **14** (6), 729–739.

Righetti, M. & Romano, G. P. 2004 Particle–fluid interactions in a plane near-wall turbulent flow. *Journal of Fluid Mechanics* **505**, 93–121.

Rosti, M. E., Ge, Z., Jain, S. S., Dodd, M. S. & Brandt, L. 2019 Droplets in homogeneous shear turbulence. *Journal of Fluid Mechanics* **876**, 962–984.

Rouson, D. W. I. & Eaton, J. K. 2001 On the preferential concentration of solid particles in turbulent channel flow. *Journal of Fluid Mechanics* **428**, 149–169.

Rycroft, C. H. 2009 VORO++ : A three-dimensional Voronoi cell library in C++. *Chaos: An Interdisciplinary Journal of Nonlinear Science* **19** (4), 041111.

Sekimoto, A., Dong, S. & Jiménez, J. 2016 Direct numerical simulation of statistically stationary and homogeneous shear turbulence and its relation to other shear flows. *Physics of Fluids* **28** (3), 035101.

Shen, X. & Warhaft, Z. 2000 The anisotropy of the small scale structure in high Reynolds number (R$\lambda\sim$1000) turbulent shear flow. *Physics of Fluids* **12** (11), 2976–2989.

Squires, K. D. & Eaton, J. K. 1991 Preferential concentration of particles by turbulence. *Physics of Fluids A: Fluid Dynamics* **3** (5), 1169–1178.

Sundaram, S. & Collins, Lance R. 1997 Collision statistics in an isotropic particle-laden turbulent suspension. Part 1. Direct numerical simulations. *Journal of Fluid Mechanics* **335**, 75–109.

Uhlmann, M. & Chouippe, A. 2017 Clustering and preferential concentration of finite-size particles in forced homogeneous-isotropic turbulence. *Journal of Fluid Mechanics* **812**, 991–1023.

Van Hinsberg, M. A. T., Ten Thije Boonkkamp, J. H. M. & Clercx, H.J.H. 2011 An efficient, second order method for the approximation of the Basset history force. *Journal of Computational Physics* **230** (4), 1465–1478.

Wang, L.-P. & Maxey, M. R. 1993 Settling velocity and concentration distribution of heavy particles in homogeneous isotropic turbulence. *Journal of Fluid Mechanics* **256**, 27–68.

Yoshimoto, H. & Goto, S. 2007 Self-similar clustering of inertial particles in homogeneous turbulence. *Journal of Fluid Mechanics* **577**, 275–286.

Yousefi, A., Niazi Ardekani, M. & Brandt, L. 2020 Modulation of turbulence by finite-size particles in statistically steady-state homogeneous shear turbulence. *Journal of Fluid Mechanics* **899**, A19.

Zamansky, R., Coletti, F., Massot, M. & Mani, A. 2016 Turbulent thermal convection driven by heated inertial particles. *Journal of Fluid Mechanics* **809**, 390–437.

A. Numerical treatment of the Basset history force

The Basset history force represents the unsteady diffusion of vorticity within the viscous boundary layer surrounding an accelerating particle and introduces a weakly singular convolution integral into the Maxey–Riley–Gatignol equation. Its inclusion is essential to recover the correct algebraic decay of the particle-fluid slip velocity and to accurately describe particle dynamics at moderate density ratios and Reynolds numbers (Jacobsen *et al.*, 2000; Prasath *et al.*, 2019; Olivieri *et al.*, 2014; Daitche, 2015).
Because the direct evaluation of the convolution over the entire particle history is computationally prohibitive, the Basset history force is computed using the second-order, memory-efficient algorithm proposed by Van Hinsberg *et al.* (2011). This method decomposes the force into a short-time window contribution and a long-time tail,

$$F_{BH}(t) = c_B \int_{t-t_{win}}^{t} K_B(t-\tau)\boldsymbol{g}(\tau)\,d\tau + c_B \int_{-\infty}^{t-t_{win}} K_{\text{tail}}(t-\tau)\boldsymbol{g}(\tau)\,d\tau = F_{BH\text{-win}}(t) + F_{BH\text{-tail}}(t), \tag{38}$$

where $c_B = \frac{3}{2}d_p^2\rho_f\sqrt{\pi\nu}$ and

$$\boldsymbol{g}(t) = \frac{d}{dt}\left(\boldsymbol{U} - \boldsymbol{U_p} + \frac{1}{6}\left(\frac{d_p}{2}\right)^2 \nabla^2\boldsymbol{U}\right). \tag{39}$$

The kernel is approximated by a composite form,

$$K(t) = \begin{cases} K_B(t), & t < t_{win}, \\ K_{\text{tail}}(t), & t \geq t_{win}, \end{cases} \qquad \lim_{t\to+\infty} K_{\text{tail}}(t) = 0, \tag{40}$$

ensuring an accurate representation of both the short-time singular behavior and the long-time decay.
The window contribution is discretized over N_w time steps using a modified trapezoidal rule that analytically treats the kernel singularity (Olivieri *et al.*, 2014):

$$\begin{aligned} F_{BH\text{-win}}(t) = {} & \frac{4}{3}c_B\boldsymbol{g}_0\sqrt{\Delta t} + \frac{4}{3}c_B\sqrt{\Delta t}\sum_{n=1}^{N_w-1}\left[(n-1)\sqrt{n-1} - 2n\sqrt{n} + (n+1)\sqrt{n+1}\right]\boldsymbol{g}_n \\ & + c_B\sqrt{\Delta t}\left[\tfrac{4}{3}(N_w-1)\sqrt{N_w-1} + \left(2 - \tfrac{4}{3}N_w\right)\sqrt{N_w}\right]\boldsymbol{g}_{N_w}, \end{aligned} \tag{41}$$

where $\boldsymbol{g}_n = \boldsymbol{g}(t - n\Delta t)$ and $\Delta t = t_{win}/N_w$.
The long-time tail, accounting for persistent memory effects, is approximated by a sum of exponentials,

$$F_{BH\text{-tail}}(t) = \sum_{i=1}^{m} a_i F_i(t), \tag{42}$$

with recursive update

$$F_{i\text{-di}}(t) = 2c_B\sqrt{et_i}\,\exp\left(-\frac{t_{win}}{2t_i}\right)\left\{\boldsymbol{g}_N\left[1 - \varphi\left(-\frac{\Delta t}{2t_i}\right)\right] + \boldsymbol{g}_{N+1}\exp\left(-\frac{\Delta t}{2t_i}\right)\left[\varphi\left(\frac{\Delta t}{2t_i}\right) - 1\right]\right\}, \tag{43}$$

$$F_{i\text{-re}}(t) = \exp\left(-\frac{\Delta t}{2t_i}\right)F_i\left(t - \Delta t\right), \tag{44}$$

where $\varphi(z) = (e^z - 1)/z$ and the coefficients $\{a_i, t_i\}_{i=1}^m$ are chosen to minimize the convolution error. Following Van Hinsberg *et al.* (2011), $m = 10$ exponential kernels are used.
The complete particle equation of motion is integrated using a semi-implicit time-stepping scheme to ensure numerical stability. A sensitivity analysis with respect to the window size and the number of exponential kernels is reported in appendix D.

B. Validation of the code

B.1. Basset integral for a prescribed convolution function

This test validates the short-time, modified-trapezoidal quadrature used to evaluate the window contribution of the Basset history force. Two closely related discretizations are compared: the formulation proposed by Van Hinsberg *et al.* (2011) (hereafter TBMH) and the variant reported by Olivieri *et al.* (2014) (hereafter TBMO). Both implementations are explicitly coded and tested. The discrete quadrature formulae implemented in the code are:

$$F_{BH\text{-win(TBMH)}}(t) = \frac{4}{3}c_B \boldsymbol{g}_0\sqrt{\Delta t} + c_B\sqrt{\Delta t}\left[\frac{N_w - \frac{4}{3}}{(N_w-1)\sqrt{N_w-1}+(N_w-\frac{3}{2})\sqrt{N_w}}\right]\boldsymbol{g}_{N_w} + c_B\sqrt{\Delta t}\sum_{n=1}^{N_w-1}\left[\frac{n+\frac{4}{3}}{(n+1)\sqrt{n+1}+(n+\frac{3}{2})\sqrt{n}} + \frac{n-\frac{4}{3}}{(n-1)\sqrt{n-1}+(n-\frac{3}{2})\sqrt{n}}\right]\boldsymbol{g}_n, \quad (45)$$

and

$$F_{BH\text{-win(TBMO)}}(t) = \frac{4}{3}c_B \boldsymbol{g}_0\sqrt{\Delta t} + \frac{4}{3}c_B\sqrt{\Delta t}\sum_{n=1}^{N_w-1}\left[(n-1)\sqrt{n-1} - 2n\sqrt{n} + (n+1)\sqrt{n+1}\right]\boldsymbol{g}_n + c_B\sqrt{\Delta t}\left[\tfrac{4}{3}(N_w-1)\sqrt{N_w-1} + \left(2-\tfrac{4}{3}N_w\right)\sqrt{N_w}\right]\boldsymbol{g}_{N_w}. \quad (46)$$

Here, $c_B = \frac{3}{2}d_p^2\rho_f\sqrt{\pi\nu}$, $\boldsymbol{g}_n = \boldsymbol{g}(t-n\Delta t)$, and $\Delta t = t_{\text{win}}/N_w$.
To assess the accuracy of the schemes, the prescribed test function $g(\tau) = \cos\tau$ is considered. The exact Basset integral can be evaluated analytically as

$$F_{BH}(t) = c_B\int_0^t \frac{\cos\tau}{\sqrt{t-\tau}}d\tau = 2c_B\int_0^{\sqrt{t}}\cos\left(t-\sigma^2\right)d\sigma = c_B\sqrt{2\pi}\left[C\left(\sqrt{2t/\pi}\right)\cos(t) + S\left(\sqrt{2t/\pi}\right)\sin(t)\right], \quad (47)$$

with $\sigma = \sqrt{t-\tau}$, and $C(t)$ and $S(t)$ denote the Fresnel cosine and sine functions, respectively (Jeffrey, 2004). The Basset integral is evaluated at $t = 50\pi$ using different numbers of quadrature points N_w, uniformly distributed over the interval $[0, 50\pi]$. In this configuration, $t_{\text{win}} = 50\pi$ and $\Delta t = 50\pi/N_w$.
The relative error of the numerical approximation is defined as

$$\varepsilon(50\pi) = \frac{\left|F_{BH\text{-win(TBMH/O)}} - F_{BH}(50\pi)\right|}{\left|F_{BH}(50\pi)\right|}. \quad (48)$$

Table 3 reports the relative errors and empirical convergence rates for both TBMH and TBMO. Both formulations exhibit clear second-order convergence over the entire resolved range. The two methods are numerically indistinguishable up to $N_w = 5\times10^5$, with differences appearing only at extreme resolutions. The slight degradation observed for TBMO at the largest N_w does not affect the overall performance, since, as discussed in appendix D, the kernel-fit error associated with the exponential-tail approximation dominates once Δt becomes sufficiently small. In all subsequent simulations, the TBMO formulation is adopted.

B.2. Sedimenting particle in a quiescent fluid

This test validates the complete window-tail implementation against the analytical solution for a particle settling under gravity in a quiescent fluid, including the Basset history force (Jacobsen *et al.*, 2000).
For a particle released from rest, the vertical component of the Maxey–Riley–Gatignol equation reduces to

$$\frac{dU}{dt} + BU + Q\int_0^t \frac{1}{\sqrt{t-s}}\frac{dU}{ds}\,ds = M, \quad (49)$$

with

$$B = \frac{9\mu}{(d_p/2)^2(2\rho_p+\rho_f)}, \quad Q = \frac{9\rho_f}{(d_p/2)(2\rho_p+\rho_f)}\sqrt{\frac{\mu}{\rho_f\pi}}, \quad M = \frac{2g(\rho_p-\rho_f)}{2\rho_p+\rho_f}.$$

The terminal velocity is $U_0 = M/B$. Introducing the nondimensional variables

$$\tau = Bt, \qquad u(\tau) = \frac{U(t)}{U_0},$$

Points N_w	Rel. error TBMH	Order TBMH	Rel. error TBMO	Order TBMO
81	$1.33 \cdot 10^{-1}$	–	$1.33 \cdot 10^{-1}$	–
243	$2.52 \cdot 10^{-2}$	1.5	$2.52 \cdot 10^{-2}$	1.5
729	$3.28 \cdot 10^{-3}$	1.9	$3.28 \cdot 10^{-3}$	1.9
2187	$3.92 \cdot 10^{-4}$	1.9	$3.92 \cdot 10^{-4}$	1.9
6561	$4.54 \cdot 10^{-5}$	2.0	$4.54 \cdot 10^{-5}$	2.0
19683	$5.15 \cdot 10^{-6}$	2.0	$5.15 \cdot 10^{-6}$	2.0
59049	$5.80 \cdot 10^{-7}$	2.0	$5.80 \cdot 10^{-7}$	2.0
177147	$6.49 \cdot 10^{-8}$	2.0	$6.50 \cdot 10^{-8}$	2.0
531441	$7.24 \cdot 10^{-9}$	2.0	$6.91 \cdot 10^{-9}$	2.0
1594323	$8.06 \cdot 10^{-10}$	2.0	$1.67 \cdot 10^{-9}$	1.3

Table 3: Relative error and empirical order of convergence for the trapezoidal-based evaluation of the Basset window integral using the Van Hinsberg *et al.* (2011) weights (TBMH) and the Olivieri *et al.* (2014) variant (TBMO).

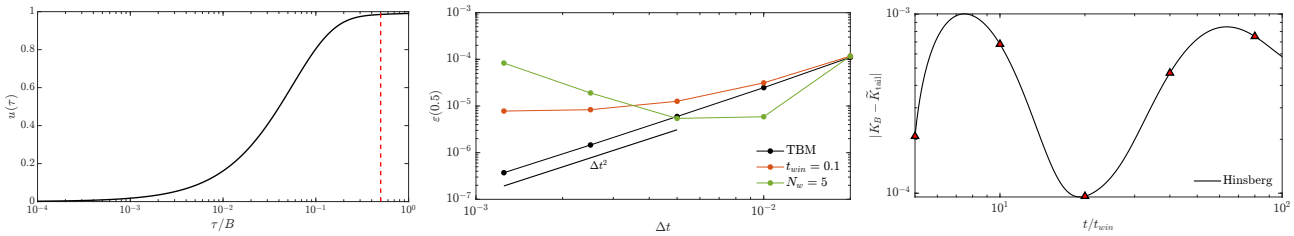
the analytical solution reads

$$u(\tau) = 1 + \frac{\sqrt{k}}{\alpha - \beta}\left(\frac{e^{\alpha\tau}\,\mathrm{erfc}(\sqrt{\alpha\tau})}{\sqrt{\alpha}} - \frac{e^{\beta\tau}\,\mathrm{erfc}(\sqrt{\beta\tau})}{\sqrt{\beta}}\right), \tag{50}$$

where $k = \pi Q^2/B$, and α, β are the two roots of $m^2 + (2-k)m + 1 = 0$.
The simulations use parameters $\rho_p/\rho_f = 1000$, $d_p/L_y = 0.01$, and $Ga = (|\rho_p/\rho_f - 1|d_p^3 g)^{1/2}/\nu \approx 316.07$, where g is the magnitude of the gravitational acceleration. Simulations are run up to $t = 1$, and the relative error of the window-tail implementation is evaluated at $t = 0.5$ against the analytical solution.
Figure 26 summarizes three numerical experiments:

1. full modified-trapezoidal integration of the entire convolution;
2. the window-tail method with fixed window length $t_{win} = 0.1$ (so $N_w = t_{win}/\Delta t$ varies as Δt changes);
3. the window-tail method with fixed number of window points $N_w = 5$ (so $t_{\mathrm{win}} = N_w \Delta t$ varies with Δt).

The full trapezoidal scheme exhibits second-order convergence. The window-tail method with fixed t_{win} plateaus once the kernel-fit error dominates. When N_w is fixed, reducing Δt shifts the evaluation point through regions of varying kernel-approximation error, resulting in non-monotonic convergence behavior.



(a) (b) (c)

Figure 26: Validation of the sedimenting-particle test. (a) Analytical solution for the vertical particle velocity (nondimensionalized). The red dashed vertical line indicates the time at which the convergence study is evaluated. (b) Relative error of the vertical velocity at the evaluation time as a function of the time step Δt, shown for three methods: the full trapezoidal method (TBM), the window-tail approach with a fixed window length $t_{win} = 0.1$, and the same approach with a fixed number of window points $N_w = 5$. (c) Zoom-in on the kernel-approximation error, $e_K(\tilde{t}) = |K_B - \widetilde{K}_{\mathrm{tail}}|$, for $m = 10$ coefficients in the Van Hinsberg *et al.* (2011) method. The markers indicate the normalized times $\tilde{t}$ corresponding to the points on the $N_w = 5$ curve in panel (b). This panel demonstrates that, when N_w is fixed and Δt decreases, the evaluation point moves into regions with larger kernel approximation errors, explaining the plateau and non-monotonic behavior observed in panel (b).

B.3. Circular trajectory in a space-dependent steady velocity field

This test, following Van Hinsberg *et al.* (2011), validates both the implementation of the material-derivative term in the added mass force and the full window-tail history-force scheme.
The prescribed particle trajectory and velocity are

$$\boldsymbol{x_p} = r\,\mathcal{R}[\mathbf{e}\,\mathrm{e}^{-i\omega t}], \qquad \boldsymbol{u_p} = -r\omega\,\mathcal{R}[i\mathbf{e}\,\mathrm{e}^{-i\omega t}], \tag{51}$$

with $\mathbf{e} = \mathbf{e_x} - i\mathbf{e_y}$. The corresponding flow velocity field satisfying the MRG equation is

$$\boldsymbol{u} = -\mathcal{R}[s(x+iy)\mathbf{e}] + 2\alpha z\,\mathbf{e_z}, \tag{52}$$

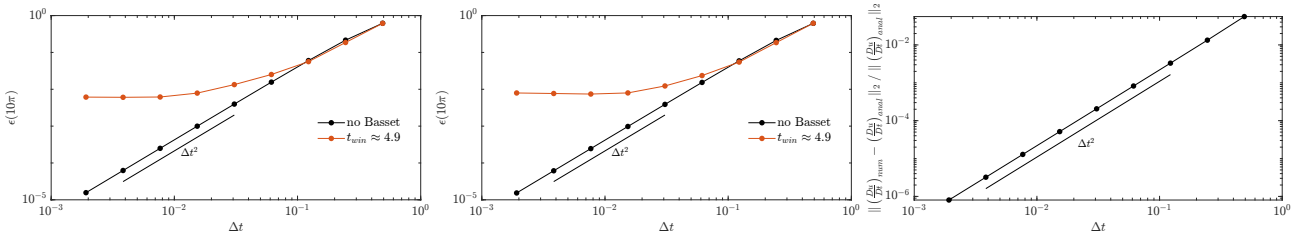
where $s = \alpha + i\beta$. A spherical particle released in the plane $z = 0$ remains confined to it by symmetry. Substituting (51) and (52) (for $z = 0$) into the MRG equation, neglecting gravity, and noting that the Faxén corrections vanish for a linear velocity field and that $\frac{D\boldsymbol{u}}{Dt} = \mathcal{R}[s^2(x+iy)\mathbf{e}]$, yields the quadratic relation for s:

$$-\omega^2\left(m_p + \frac{1}{2}m_f\right) = 6\pi\frac{d_p}{2}\mu\,(-s+i\omega) + \frac{3}{2}m_f s^2 + c_B\sqrt{\frac{\omega\pi}{2}}(\omega+is)(1+i). \tag{53}$$

Two solutions exist for s, but only one corresponds to a physically admissible particle trajectory; the other is discarded.
Five revolutions are simulated. First, the material-derivative term is validated by excluding the Basset history force; second, the full window-tail scheme is tested. The global relative error after five periods is defined as

$$\epsilon(5T) = \frac{\|\boldsymbol{x}(5T) - \boldsymbol{x}(0)\|_2}{\|\boldsymbol{x}(0)\|_2}, \tag{54}$$

where $\boldsymbol{x} = (x_1, x_2)$ denotes either the particle position or velocity in the plane $z = 0$.
Figure 27 shows second-order convergence for the no-Basset case and confirms Δt^2 accuracy for the material derivative. For the window-tail scheme the error initially follows second-order behavior but then plateaus when the kernel-approximation error of the exponential tail dominates, as discussed in appendix D.



(a) (b) (c)

Figure 27: Validation of the circular-trajectory test. (a) Relative error of the particle position and (b) relative error of the particle velocity at $t = 10\pi$ as a function of the time step Δt, shown for two cases: without the Basset term, and using the Van Hinsberg *et al.* (2011) window-tail method with fixed $t_{win} \approx 4.9$. The no-Basset case shows second-order convergence, while the window-tail solution initially follows this trend until the kernel-fit error becomes dominant, causing the error to plateau. (c) (c) Mean relative error of the computed material derivative $D\boldsymbol{u}/Dt$ compared to the analytical value as a function of Δt.

C. Derivation of the Maxey–Riley–Gatignol equation for homogeneous shear turbulence

This appendix presents a detailed derivation of the Maxey–Riley–Gatignol equation specialized for inertial particles in a homogeneous shear turbulence configuration. The goal is to show how the governing equation naturally splits into contributions from turbulent fluctuations and explicit mean shear coupling.
We start from the complete MRG equations for a small spherical particle:

$$\begin{aligned}\rho_p V_p \frac{d\boldsymbol{U_p}}{dt} = &\underbrace{3\pi d_p \rho_f \nu \left(\boldsymbol{U} - \boldsymbol{U_p} + \frac{1}{6}\left(\frac{d_p}{2}\right)^2 \nabla^2 \boldsymbol{U} \right)}_{\textit{Stokes drag}} + \underbrace{\rho_f V_p \frac{D\boldsymbol{U}}{Dt}}_{\textit{Pressure gradient}} \\ &+ \underbrace{\frac{\rho_f V_p}{2}\left(\frac{D\boldsymbol{U}}{Dt} - \frac{d\boldsymbol{U_p}}{dt} + \frac{1}{10}\left(\frac{d_p}{2}\right)^2 \frac{d}{dt}(\nabla^2 \boldsymbol{U}) \right)}_{\textit{Added mass}} \\ &+ \underbrace{\frac{3}{2} d_p^2 \rho_f \sqrt{\pi\nu} \int_{-\infty}^{t} K_B(t-\tau) \left(\frac{d\boldsymbol{U}}{d\tau} - \frac{d\boldsymbol{U_p}}{d\tau} + \frac{1}{6}\left(\frac{d_p}{2}\right)^2 \frac{d}{d\tau}\left(\nabla^2 \boldsymbol{U}\right) \right) d\tau}_{\textit{Basset history force}}, \\ \frac{d\boldsymbol{x_p}}{dt} = &\boldsymbol{U_p},\end{aligned} \tag{55}$$

where $\boldsymbol{U}$ is the undisturbed fluid velocity and $\boldsymbol{U_p}$ the particle velocity, d_p is the particle diameter, ρ_f and ρ_p are the fluid and particle densities, and K_B is the Basset kernel.

C.1. Flow decomposition for HST

In homogeneous shear turbulence, the fluid and particle velocities can be decomposed into a mean shear and a turbulent fluctuation:

$$\boldsymbol{U}(\boldsymbol{x}, t) = \boldsymbol{U_0}(z) + \boldsymbol{u}(\boldsymbol{x}, t), \qquad \boldsymbol{U_p} = \boldsymbol{U_0}(\boldsymbol{x_p}(t)) + \boldsymbol{u_p}(t),$$

where S is the mean shear rate, $\boldsymbol{u}$ is the fluid fluctuation, and $\boldsymbol{u_p}$ is the fluctuating particle velocity. The material derivative along the fluid and along the particle trajectories are

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \boldsymbol{U} \cdot \nabla, \qquad \frac{d}{dt} = \frac{\partial}{\partial t} + \boldsymbol{U_p} \cdot \nabla.$$

Since the mean flow is linear and steady, $\boldsymbol{U_0} = Sz\,\hat{\boldsymbol{x}}$, we have $\partial_t \boldsymbol{U_0} = 0$ and $\nabla^2 \boldsymbol{U_0} = 0$. This simplification allows each force term to be decomposed systematically.

C.2. Component-wise Decomposition

C.2.1 Particle inertia

The particle acceleration splits into mean shear and fluctuation contributions:

$$\rho_p V_p \frac{d\boldsymbol{U_p}}{dt} = \rho_p V_p \left(\frac{d\boldsymbol{U_0}}{dt} + \frac{d\boldsymbol{u_p}}{dt} \right). \tag{56}$$

For the mean shear term,

$$\frac{d\boldsymbol{U_0}}{dt} = \cancel{\frac{\partial \boldsymbol{U_0}}{\partial t}} + (\boldsymbol{U_p} \cdot \nabla)(Sz\,\hat{\boldsymbol{x}}) = \left[(Sz_p + u_p)\frac{\partial}{\partial x} + v_p \frac{\partial}{\partial y} + w_p \frac{\partial}{\partial z} \right](Sz\,\hat{\boldsymbol{x}}) = Sw_p\,\hat{\boldsymbol{x}}, \tag{57}$$

so the total inertia is

$$\rho_p V_p \frac{d\boldsymbol{U_p}}{dt} = \rho_p V_p \left(Sw_p \hat{\boldsymbol{x}} + \frac{d\boldsymbol{u_p}}{dt} \right), \tag{58}$$

highlighting that vertical particle motion w_p couples directly to streamwise acceleration, while fluctuations evolve independently.

C.2.2 Stokes Drag and Basset history force

Since $\nabla^2 \boldsymbol{U_0} = 0$ and the mean flow cancels in the particle slip, these forces act only on the fluctuations:

$$F_{SD} = 3\pi d_p \rho_f \nu \left(\boldsymbol{u} - \boldsymbol{u_p} + \frac{1}{6} \left(\frac{d_p}{2} \right)^2 \nabla^2 \boldsymbol{u} \right), \tag{59}$$

$$F_{BH} = \frac{3}{2} d_p^2 \rho_f \sqrt{\pi \nu} \int_{-\infty}^{t} K_B(t-\tau) \frac{d}{d\tau} \left(\boldsymbol{u} - \boldsymbol{u_p} + \frac{1}{6} \left(\frac{d_p}{2} \right)^2 \nabla^2 \boldsymbol{u} \right) d\tau. \tag{60}$$

C.2.3 Pressure gradient

The pressure gradient term depends on the material derivative of the fluid velocity at the particle position:

$$F_{PG} = \rho_f V_p \frac{D\boldsymbol{U}}{Dt} = \rho_f V_p \left(\frac{D\boldsymbol{u}}{Dt} + (\boldsymbol{U} \cdot \nabla) (Sz\, \hat{\boldsymbol{x}}) \right) = \rho_f V_p \left(\frac{D\boldsymbol{u}}{Dt} + Sw\, \hat{\boldsymbol{x}} \right), \tag{61}$$

where w is the wall-normal fluid velocity. This shows an explicit coupling between the mean shear and the wall-normal velocity, alongside the fluctuating acceleration.

C.2.4 Added mass

The added-mass force accounts for the fluid inertia accelerated with the particle. Using the same decomposition for fluid and particle velocities and recalling that $\nabla^2 \boldsymbol{U}_0 = 0$, it reduces to

$$F_{AM} = \frac{\rho_f V_p}{2} \left(\frac{D\boldsymbol{u}}{Dt} - \frac{d\boldsymbol{u}_p}{dt} + \frac{1}{10} \left(\frac{d_p}{2} \right)^2 \frac{d}{dt} \nabla^2 \boldsymbol{u} + S(w - w_p)\, \hat{\boldsymbol{x}} \right). \tag{62}$$

It acts both on the relative acceleration between fluid and particle and introduces a shear-induced streamwise contribution proportional to the relative wall-normal velocity.

C.3. Final MRG equation for HST

Combining all contributions gives the specialized form:

$$\begin{aligned}
\rho_p V_p \frac{d\boldsymbol{u_p}}{dt} = & \underbrace{3\pi d_p \rho_f \nu \left(\boldsymbol{u} - \boldsymbol{u_p} + \frac{1}{6} \left(\frac{d_p}{2} \right)^2 \nabla^2 \boldsymbol{u} \right)}_{\textit{Fluctuating Stokes drag}} + \underbrace{\rho_f V_p \frac{D\boldsymbol{u}}{Dt}}_{\textit{Fluctuating pressure gradient}} \\
& + \underbrace{\frac{\rho_f V_p}{2} \left(\frac{D\boldsymbol{u}}{Dt} - \frac{d\boldsymbol{u_p}}{dt} + \frac{1}{10} \left(\frac{d_p}{2} \right)^2 \frac{d}{dt} \nabla^2 \boldsymbol{u} \right)}_{\textit{Fluctuating added mass}} \\
& + \underbrace{\frac{3}{2} d_p^2 \rho_f \sqrt{\pi \nu} \int_{-\infty}^{t} K_B(t-\tau) \left(\frac{d\boldsymbol{u}}{d\tau} - \frac{d\boldsymbol{u_p}}{d\tau} + \frac{1}{6} \left(\frac{d_p}{2} \right)^2 \frac{d}{d\tau} \nabla^2 \boldsymbol{u} \right) d\tau}_{\textit{Fluctuating Basset history force}} \\
& + \underbrace{S V_p \left(\frac{3}{2} \rho_f w - \left(\rho_p + \frac{\rho_f}{2} \right) w_p \right) \hat{\boldsymbol{x}}}_{\textit{Mean shear coupling}}, \\
\frac{d\boldsymbol{x_p}}{dt} = & \boldsymbol{u_p} + S z_p\, \hat{\boldsymbol{x}}.
\end{aligned} \tag{63}$$

In this form, each term clearly separates turbulence-driven forces (acting on the fluctuating velocity) from the mean shear effect, which manifests as a direct coupling between vertical motion and streamwise acceleration.

D. Sensitivity analysis of the exponential-tail approximation for the Basset history time kernel

The exponential-tail approximation represents the long-time contribution of the Basset kernel as a finite sum of exponentially decaying modes,

$$K_{\text{tail}}(t) = \sum_{i=1}^{m} a_i K_i(t), \qquad K_i(t) = \sqrt{\frac{\mathrm{e}}{t_i}} \exp\left(-\frac{t}{2t_i}\right). \tag{64}$$

This approach reproduces the algebraic decay of the exact kernel using a small number of recursively updated modes, thereby avoiding the storage of the full particle history.
The effect of the number of exponentials m is assessed using the convolution of a sine function with the Basset kernel,

$$I_{\text{ref}}(t) = \int_0^t \frac{\sin\tau}{\sqrt{t-\tau}}\, d\tau = \sqrt{2\pi}\left[C\left(\sqrt{2t/\pi}\right)\sin(t) - S\left(\sqrt{2t/\pi}\right)\cos(t)\right], \tag{65}$$

where C and S denote the Fresnel cosine and sine integrals, respectively (Jeffrey, 2004). This problem corresponds to a harmonically oscillating particle in a quiescent fluid and is sensitive to both amplitude and phase errors in the history force.
Errors are evaluated at $t = 10$ using a window length $t_{\text{win}} = 1$. The relative error is defined as

$$E(10) = \frac{|I_{\text{app}}(10) - I_{\text{ref}}(10)|}{|I_{\text{ref}}(10)|}, \tag{66}$$

where I_{app} combines explicit window integration with the exponential-tail approximation.
Following Van Hinsberg *et al.* (2011), time and kernel are rescaled as

$$\widetilde{K}_{\text{tail}}(\tilde{t}) = \frac{K_{\text{tail}}(t)}{K_B(t_{win})}, \qquad \widetilde{K}_B(\tilde{t}) = \frac{K_B(t)}{K_B(t_{win})} = K_B(\tilde{t}), \qquad \tilde{t} = \frac{t}{t_{win}}, \tag{67}$$

which allows the kernel-approximation error to be expressed compactly as $e_K(\tilde{t}; a_i, \tilde{t}_i) = K_B(\tilde{t}) - \widetilde{K}_{\text{tail}}(\tilde{t}; a_i, \tilde{t}_i)$.
Figure 28a compares the full modified trapezoidal method (TBM), the van Hinsberg parameterization ($m = 10$), and the optimized coefficient sets of Casas *et al.* (2018) for $m = 1, \ldots, 10$ (obtained by minimizing the I_{2t} cost functional defined in Casas *et al.* (2018)).
The TBM exhibits second-order convergence with Δt. The window-tail schemes initially follow the same slope but reach an error plateau as Δt decreases, reflecting the dominance of the kernel-approximation error. Notably, the Casas set with $m = 7$ achieves accuracy comparable to the van Hinsberg set with $m = 10$.
The kernel-fit error e_K as a function of normalized time $\tilde{t} = t/t_{\text{win}}$ (figure 28b) shows two main trends: *(i)* e_K decays with increasing $\tilde{t}$ for any fixed m, ensuring diminishing influence of the approximated tail at late times; and *(ii)* for small-to-moderate $\tilde{t}$, e_K is highly sensitive to m, with additional exponentials providing orders-of-magnitude error reduction, while differences vanish at large $\tilde{t}$.

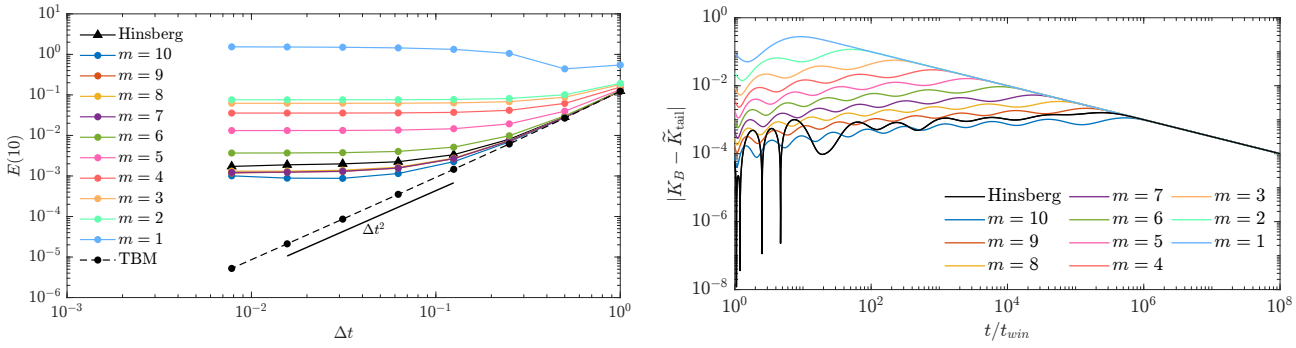


Figure 28: (a) Relative error $E(10)$ versus time step Δt for the sine-kernel convolution test: full trapezoidal method (TBM), van Hinsberg parameterization, and Casas-optimized coefficient sets ($m = 1, \ldots, 10$). (b) Kernel-fit error e_K versus normalized time $\tilde{t} = t/t_{win}$ for the same parameterizations.

Abstract in lingua italiana

Vengono eseguite simulazioni numeriche dirette della dinamica delle particelle inerziali in turbolenza di taglio omogenea per studiare il raggruppamento delle particelle e i meccanismi alla base della concentrazione preferenziale. Il moto delle particelle è descritto dall'equazione completa di Maxey–Riley–Gatignol, che include la forza di Basset, consentendo una valutazione esplicita degli effetti di memoria. Le simulazioni sono condotte per numeri di Stokes $St = [0.1, 0.5, 1]$ e rapporti di densità particella-fluido $\rho_p/\rho_f = [10, 100]$, consentendo un'esplorazione sistematica del comportamento dipendente dall'inerzia e dalla densità.
L'intensità del raggruppamento aumenta con l'inerzia delle particelle ed è localizzata prevalentemente su scale dissipative, dove le particelle campionano preferenzialmente regioni dominate dalla deformazione evitando strutture vorticose intense. Il taglio medio induce una pronunciata anisotropia nella distribuzione delle particelle e nell'orientamento dei cluster, ma non inibisce il raggruppamento o il suo carattere multiscala.
L'analisi di Voronoï rivela distribuzioni di dimensioni dei cluster autosimili che mostrano una scala di potenza e una concentrazione interna di particelle quasi costante in un ampio intervallo di dimensioni dei cluster, indicando dinamiche di aggregazione invarianti in scala. La morfologia dei cluster è caratterizzata da forme allungate e mostra un alto grado di universalità, con una dipendenza solo debole dal numero di Stokes e dal rapporto di densità, in linea con i risultati precedenti nella turbolenza isotropa omogenea. L'inclusione della forza di Basset riduce sistematicamente l'intensità del raggruppamento, appiana la geometria dei cluster e altera la penetrazione delle particelle nelle regioni ad alta enstrofia.
Questi risultati dimostrano che il raggruppamento delle particelle inerziali nella turbolenza di taglio omogenea deriva dall'interazione tra dinamiche su scala dissipativa, anisotropia indotta dal taglio ed effetti di memoria non locali, che portano a un'organizzazione delle particelle robusta ma fortemente strutturata su tutte le scale.

Parole chiave: Turbolenza, Particelle, Forza di Basset, Clustering, Campionamento preferenziale

Acknowledgements

Desidero innanzitutto ringraziare Alessandro Chiarini e il Prof. Maurizio Quadrio per avermi dato l'opportunità di affrontare questo lavoro di tesi. Un ringraziamento speciale va ad Alessandro Chiarini per il costante supporto, i preziosi consigli e il tempo che mi ha dedicato, non solo dal punto di vista accademico, ma anche umano.
Ci tengo inoltre a ringraziare gli amici e compagni di università con cui ho condiviso momenti indimenticabili in questi anni: dalle colazioni in mensa ai progetti, fino alle gite a Verbania. Alberto, Beatrice, Daniele, Federico, Francesco, Giacomo, Giorgia, Mattia, Paolo, Roberto, Silvia e Tommaso: grazie per aver reso questo percorso ancora più bello.
Un pensiero speciale va anche agli amici di casa che, anche se non ci vediamo spesso, porto sempre nel cuore: Federico, Giacomo, Marco, Margherita, Luca e Sergiu.
Infine, il grazie più grande va ai miei genitori e alle mie sorelle, Irene e Silvia, per il sostegno costante e per aver sempre creduto in me. Per voi, un semplice grazie non sarà mai abbastanza.